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SA-STD-103
29 May 1961

MILITARY STANDARD

PATCHING OF WOOD STOCKS

FOR THE 7.62MM, M14 AND M14E2 RIFLES

FSC 1005

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DEPARTMENT OF THE ARMY

U.S. ARMY MATERIEL COMMAND

WASHINGTON, D.C.

Patching of Wood Stocks

for the 7.62 MM, M14 and M14E2 Rifles

MIL-STD-127OA (WC)

1. This standard has been approved by the Department of the Army and is published to provide standards for patching of walnut and birch wood stocks for the 7.62 MM, M14 and M14E2 rifles.

2. Recommended corrections, additions or deletions should be addressed to the Commanding General, U.S. Army Weapons Command, ATTN: AMSWE-RDT, Rock Island, Ill. 61201.

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FOREWORD

Because of the decline in quantity of black walnut wood available for manufacture into rifle stocks and the increase in cost of wood blanks for this purpose, it has become necessary to authorize the use of birch as an alternate material and to permit repair of defects in wood stocks by patching and filling.

The standards contained herein have been developed to assure that uniform materials and procedures are utilized by all manufacturers, and that the size and location of patches and fills are such that they will not reduce the military effectiveness of the 7.62 MM, M14 and M14E2 Rifles.

Photographic illustrations have been included which show examples of defects before and after repair, and also examples of unacceptable repairs.

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1. SCOPE

1.1 This document establishes standard materials and procedures for patching of wood stocks for the 7.62 MM, M14 and M14E2 rifles, and limits the size and location of defects which may be repaired by the application of wood patches and plastic fillers.

1.1.1 The type of defects covered herein are borer holes, checks, open defects, edge splits, knotholes and workmanship.

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

MMM–A-181-— Adhesive, Room-Temperature and Interrnediate-Temperature Setting Resin (Phenol, Resorcinol, and Melamine Base).

3. DEFINITIONS

3.1 For the purpose of this standard the following definitions shall apply.

3.1.1 *Borer holes.* Voids made by wood-boring insects or worms.

3.1.2 *Checks.* A partial separation of wood fibers running parallel to the grain of the wood, and caused chiefly by stresses produced during kiln drying or seasoning.

3.1.3 *Chip pocket.* A cavity in the otherwise sound surface of wood caused by improper handling or machining.

3.1.4 *Edge splits.* Wedge-shaped openings caused by splitting of the wood during machining or handling.

3.1.5 *Height dimension.* The vertical linear distance of the patch in a plane at right angles to the longitudinal axis of the stock when the stock is in a horizontal position as shown by the side view in figure 1.

3.1.6 *Knots.* Cross section of a branch or limb whose grain usually runs at right angles to that of the piece in which it is found.

3.1.7 *Knotholes.* Voids produced by the separation of knots from the wood in which they were originally embedded.

3.1.8 *Length dimension.* The linear distance of the patch in a plane parallel to the longitudinal axis of the stock.

3.1.9 *Open defects.* Open checks, knotholes, loose knots, chip pockets, and other defects interrupting the smooth continuity of the surface.

3.1.10 *Patch.* An insert of sound wood for replacing defects.

3.1.11 *Plastic filler.* A fiber and resin aggregate with quick-drying agent used to fill small open defects.

3.1.12 *Router patch.* A patch with parallel sides and rounded corners.

3.1.13 *Wedge patch.* A rectangular shaped patch with two surfaces intersecting to form a knife edge.

3.1.14 *Width dimension. The* linear distance of the patch in a plane at right angles to the height dimension,

4. GENERAL REQUIREMENTS

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4.1 Patching and plastic filling of defects in M14 and M14E2 rifle stocks shall be accomplished by the contractor in accordance with the requirements contained herein.

4.2 Patching of stock areas other than those permitted by this standard (4.4), or application of more than two patches to an individual stock shall require prior approval of the procuring agency.

4.2.1 When two patches are applied to an individual stock, not more than one patch shall be permitted in the area identified as the butt area and pistol grip area in figure 1, and not more than one patch shall be permitted in the remainder of the stock.

4.2.1.1 For the purpose of this standard, the patch illustrated in figure 6A shall be counted as one patch.

4.3 Photographs depicting examples of acceptable and not acceptable patches and fills are contained herein to illustrate general and detailed requirements. The contrast between the patch and the stock has been emphasized in some of these photographs to more clearly define the patched area: therefore, they should not be used as standards for color match (see 5.3.4.2).

4.4 General patching criteria by areas (see figures 1 and 1A).

4.4.1 *Butt area.* No patches permitted to the left of a plane through the vertical centerline of the stock except for an area extending one inch forward from the rear surface and *as* indicated in figure 6A and table I. Figures 2A through 5A and 7A illustrate typical patches permitted on the remainder of the butt area. (See figures 21 through 26 for typical examples of not acceptable patches in the butt area.)

4.4.2 *Pistol grip area. No* patching permitted.

4.4.3 *Receiver opening area.* Figures 8A, 9A, 13A, 14A, and 16 illustrate typical patches permitted in this area. Patches in other portions of the receiver opening area are not permitted without prior approval by the procuring agency (see 4.2).

4.4.4 *Forearm area.* Figures l0A, 11A, and 12A illustrate typical patches permitted in this area. Wedge patches are permitted (see 5.3.4.1). For the M14E2 rifle stock, no patch is permitted on the bottom surface of the forearm area (see figure 15B). (See figure 27 for typical example of not acceptable patch in the forearm area.)

4.4.5 *Front band area.* Figure 15A illustrates patch permitted in this area. No other patches, except as illustrated in figures 11A and 12A, are permitted without prior approval of the procuring agency (see 4.2). (See figure 28 for typical not acceptable patch in the front band area.)

5. DETAIL REQUIREMENTS

5.1 Materials.

5.1.1 The adhesive used for assembly gluing of patches shall conform to MMM–A–181.

5.l.2 The plastic filler material shall be wood flour and cellulosic resin binder with suitable solvent capable of rapid drying when exposed to air. The color of the plastic filler shall be walnut; however, a clear plastic filler may be used when the application is for birch stocks.

5.2 Storage and preparation of adhesive.

5.2.1 *Storage and preparation of adhesive.* The storage and preparation of adhesive shall be in accordance with the manufacturer's printed instructions.

5.3 Patches.

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5.3.1 Patches shall not be applied to the stock as indicated in 4.4.

5.3.2 The size of the patch shall not exceed the limits specified in table I.

TABLE I. *Maximum Patch Size*

Visual standard figure No.	Maximum patch dimension (in.)		
	Length	Height	Width
2A	1-3/4	3/4	5/8
3A	1	7/16	1/2
4A	3/4	2-1/4	5/8
5A	2-3/4	1-7/8	7/16
6A	2	3/4	5/16
7A	7/8	3/4	1/4
8A	1-5/8	5/32	1/8
9A	1	1/8	3/16
10A	1-1/2	3/4	1/8
11A	4	1/4	3/16
12A	2-5/8	3/16	1/4
13A	1	1/4	3/8
14A	3/4	3/16	5/32
15A	(See figs. 15A and 15B)	(See fig. 15A)	Resultant
16	(See fig. 16)	3/16	7/8

Note. Maximum size patch, either router or wedge. in the forearm area (see figure 1) shall not exceed 8 inches in length or 1/8 inch in heigh, (Width varies dependent upon stock contour.)

5.3.4 *Preparation of parts.*

5.3.4.1 The patch to be bonded shall be a router patch of the same material as the stock being repaired and shall be machined or otherwise prepared and prefitted to insure a good mechanical fit. The patch cavity in the stock shall be machined to accept the router patch. Wedge patches shall not be used except in the forearm area.

5.3.4.2 The grain structure and color of the patch shall resemble the stock area surrounding the cavity in which the patch is to be bonded. Unless otherwise specified herein, the direction of the grain of the assembled patch shall be similar to that of the surrounding wood.

5.3.4.3 Prior to bonding, patch and stock surfaces to be glued shall be thoroughly cleaned of all dust, dirt, w-ax, grease or other foreign material.

5.3.5 *Application and use of adhesive.* The application and use of adhesive in bonding the patch to the stock shall be in accordance with the manufacturer's instructions.

5.3.6 *Fins/ patch surface.* The surface of the bonded patch after final surfacing or machining shall be on the same plane as the surrounding stock surface. Sunken areas shall not be permitted. Sanding where necessary shall be done in the direction of the grain of the wood. Staining shall be in accordance with applicable stock drawing.

5.4 Fillers.

5.4.1 Plastic fillers may be used to repair small shallow checks, knotholes, borer holes and chip pockets (see figures 17, 18, 19 and 20). The defect shall be cleaned of all loose and decayed wood and other foreign material before being filled with the plastic material.

5.4.2 The plastic filler being inserted into the defect cavity shall have the same pliability and texture as from a freshly opened

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container. No dried-out or flaky plastic material shall be used.

5.4.3 The defect cavity shall be amply filled with the plastic material to provide for any shrinkage inherent in the material. The surface of the fill after final surfacing or machining shall be on the same plane as the surrounding stock surface. Sunken areas shall not be permitted. Sanding where necessary shall be done in the direction of the grain of the wood.

5.4.4 The size of defects repaired by the application of plastic filler shall not exceed the limits specified in table II. The size indicated includes the clean-out area of the defect.

TABLE II. *Maximum Filler Size*

Defect	Maximum filler dimension (in)
Checks	1/16 wide, 1/2 long and 1/8 deep
Knotholes	1/4 diameter, 1/8 deep
Borer holes	1/16 diameter, 1/8 deep
Chip pockets	1/8 wide, 1/4 long, 1/8 deep

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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Preparing activity:

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(Springfield Armory)

Project No. 1005-A291

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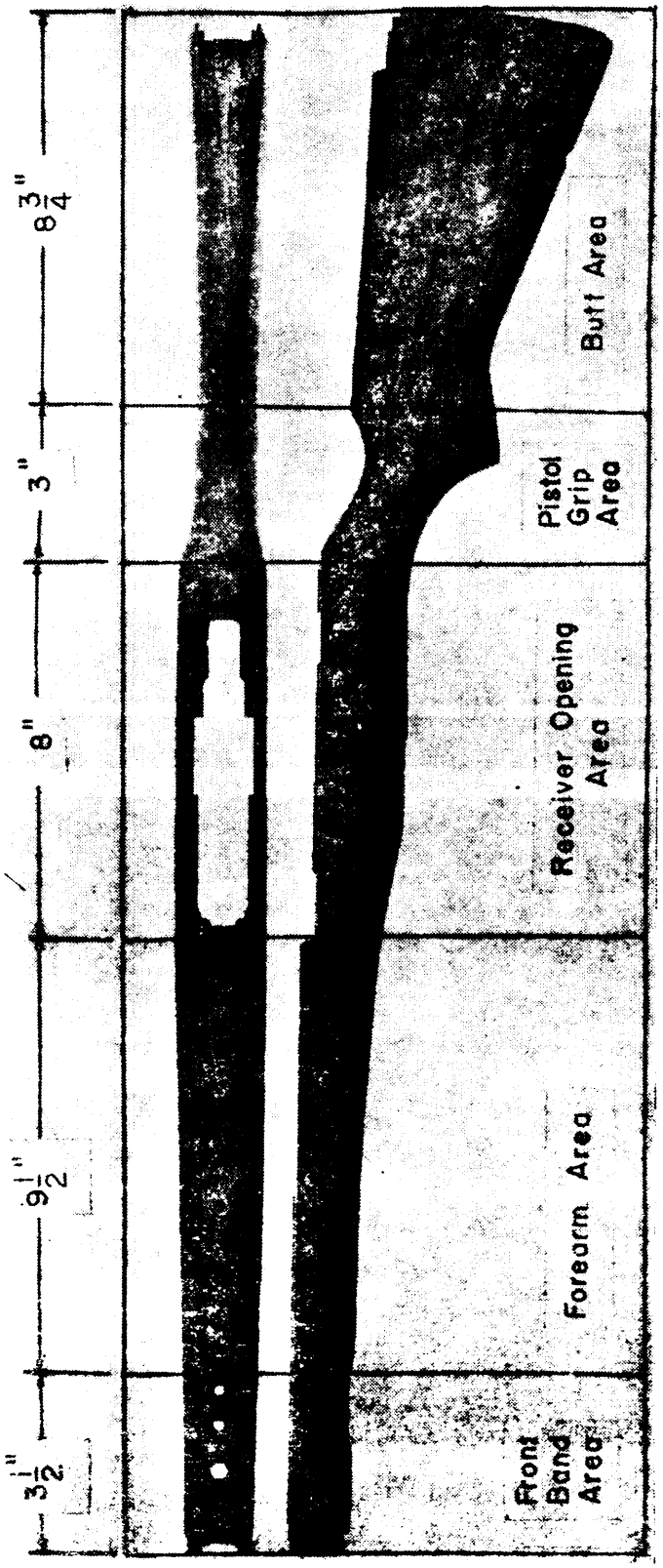


FIGURE 1. *Stock areas.*

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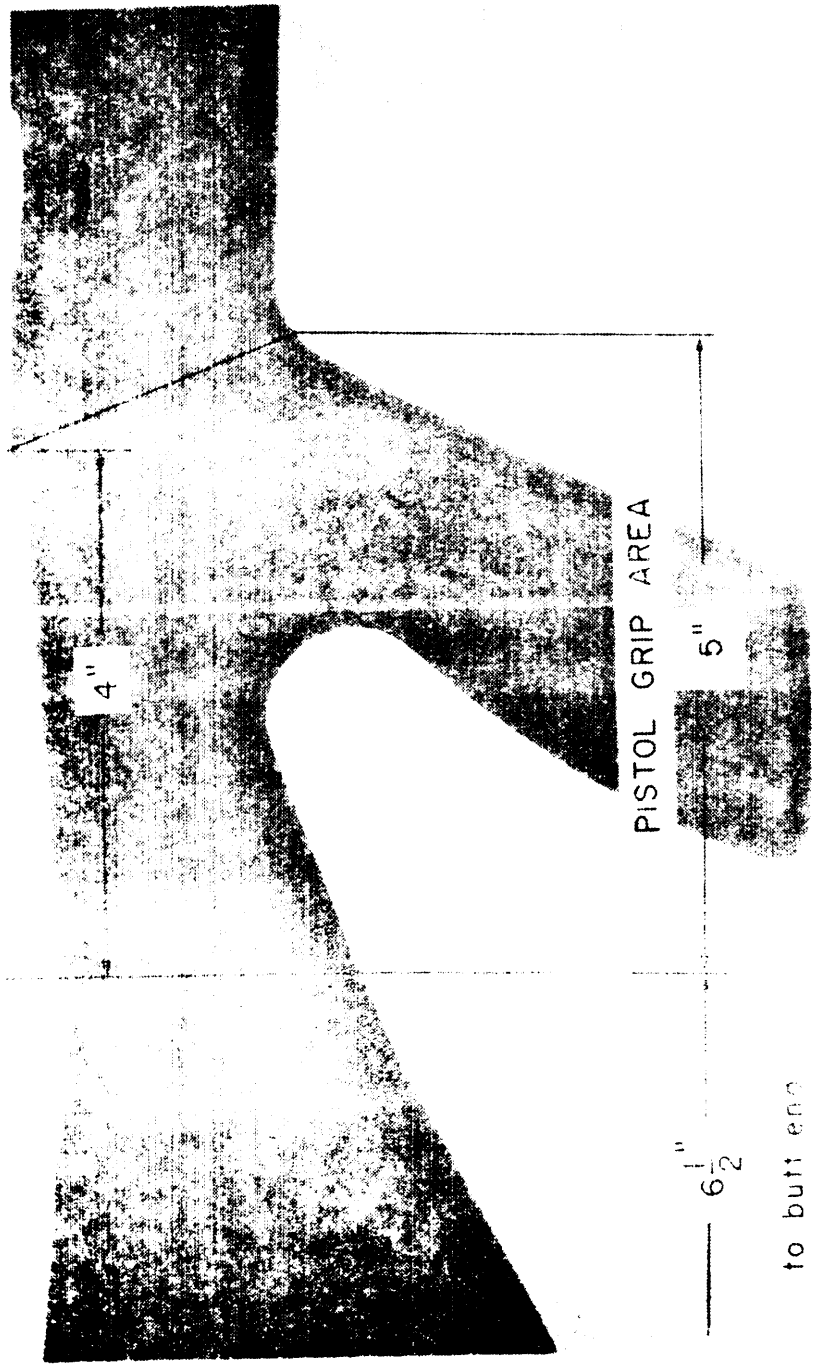


Fig. 1A. Pistol grip area of .22 rifle stock.

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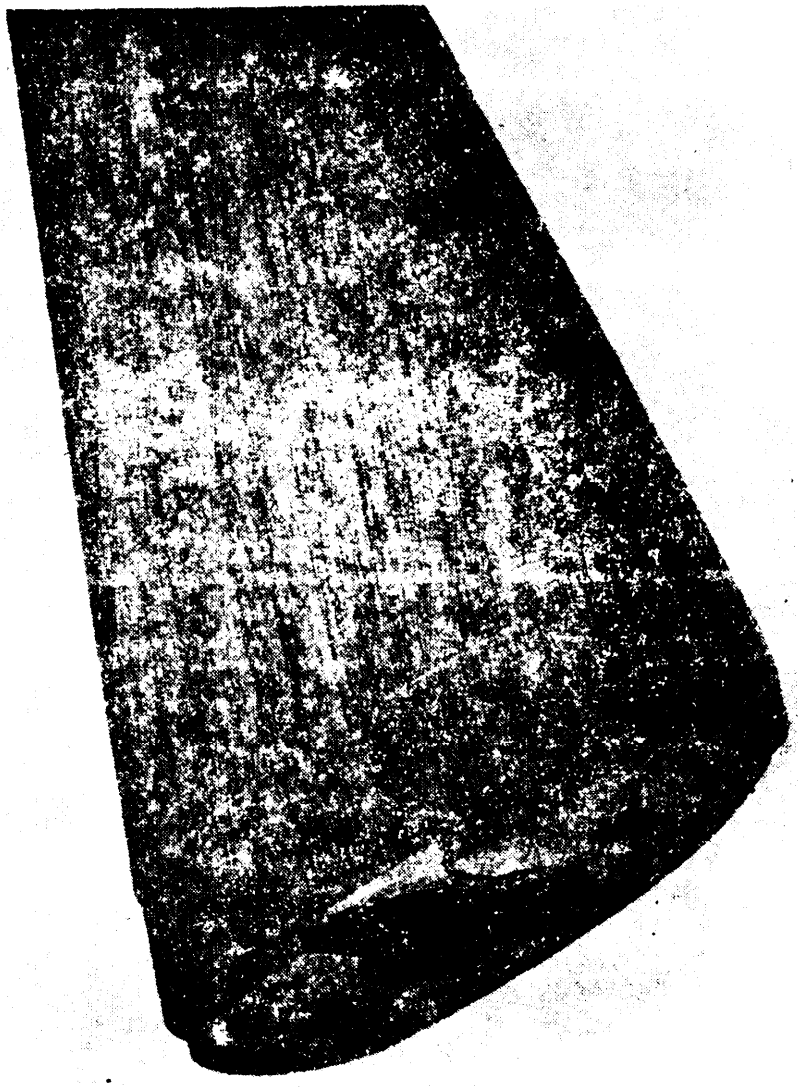


FIGURE 2. *Edge split defect.*

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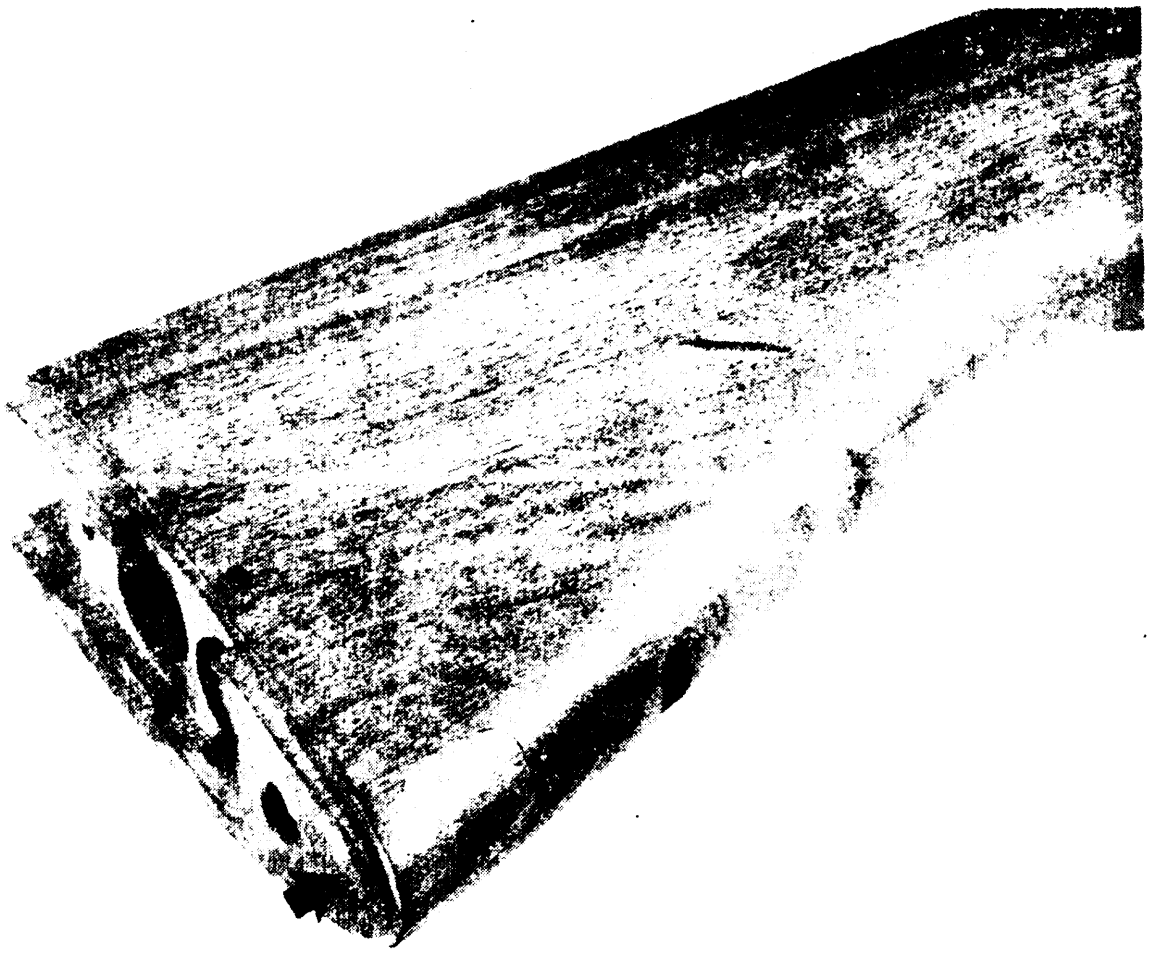


FIGURE 2A. *Edge split defect, patched.*

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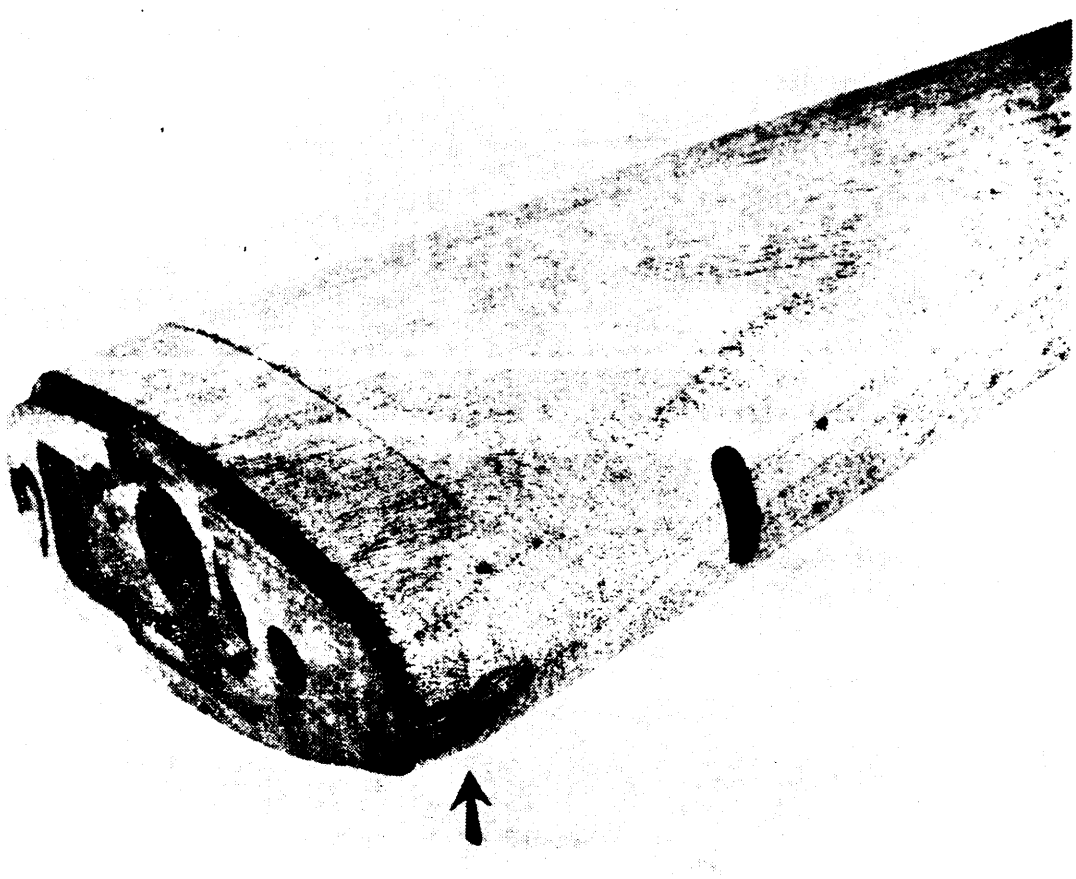


FIGURE 3. *Split defect.*

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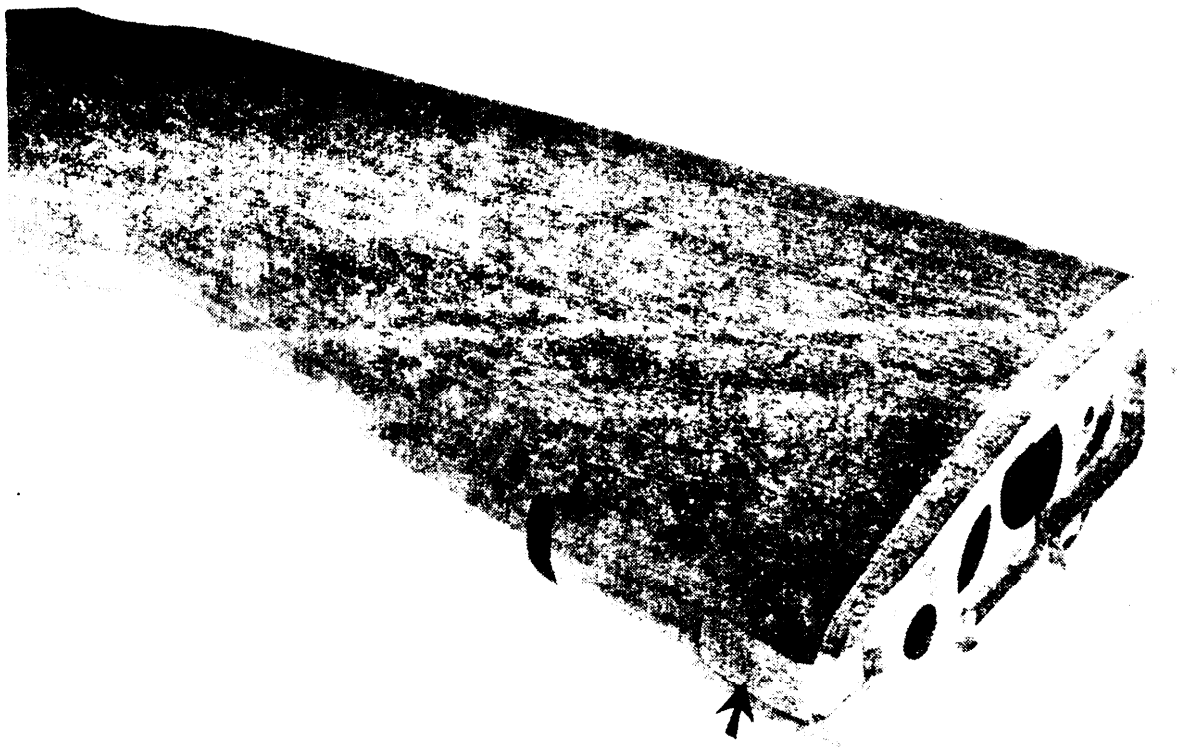


FIGURE 3A. *Split defect, patched.*

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FIGURE 4. *Knothole defect.*

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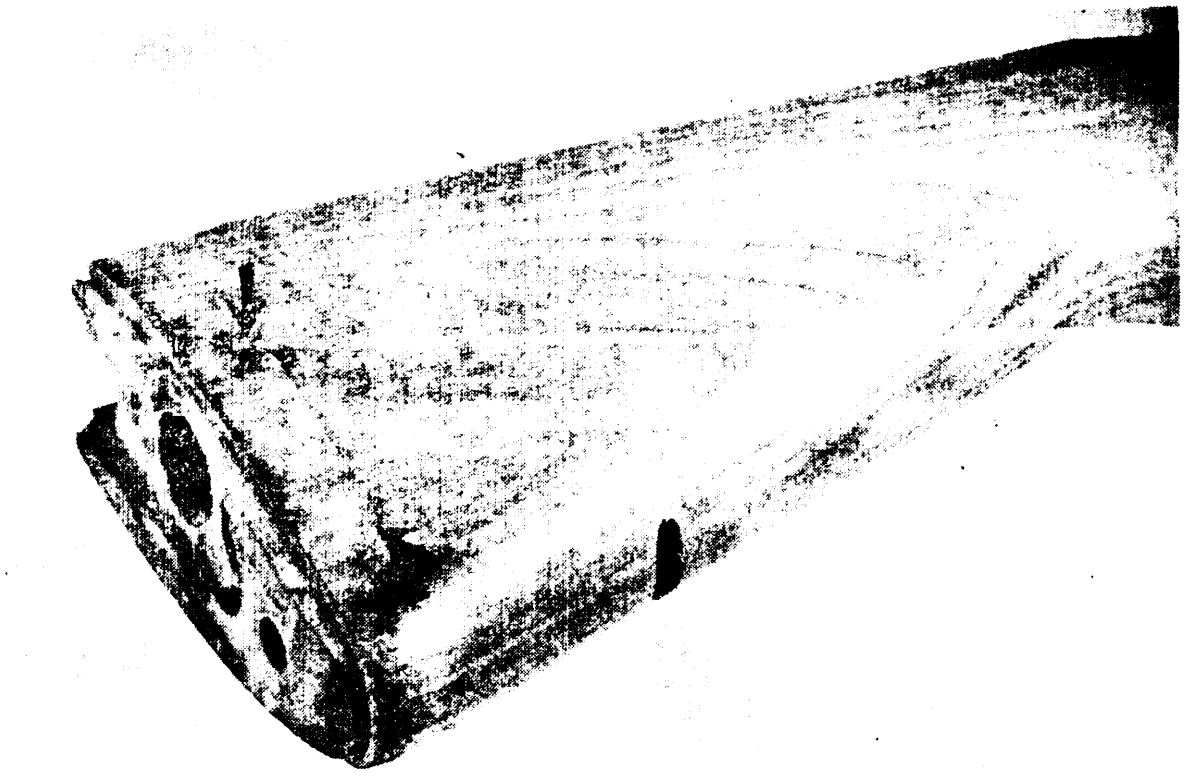


FIGURE 4A. *Knothole defect, patched.*

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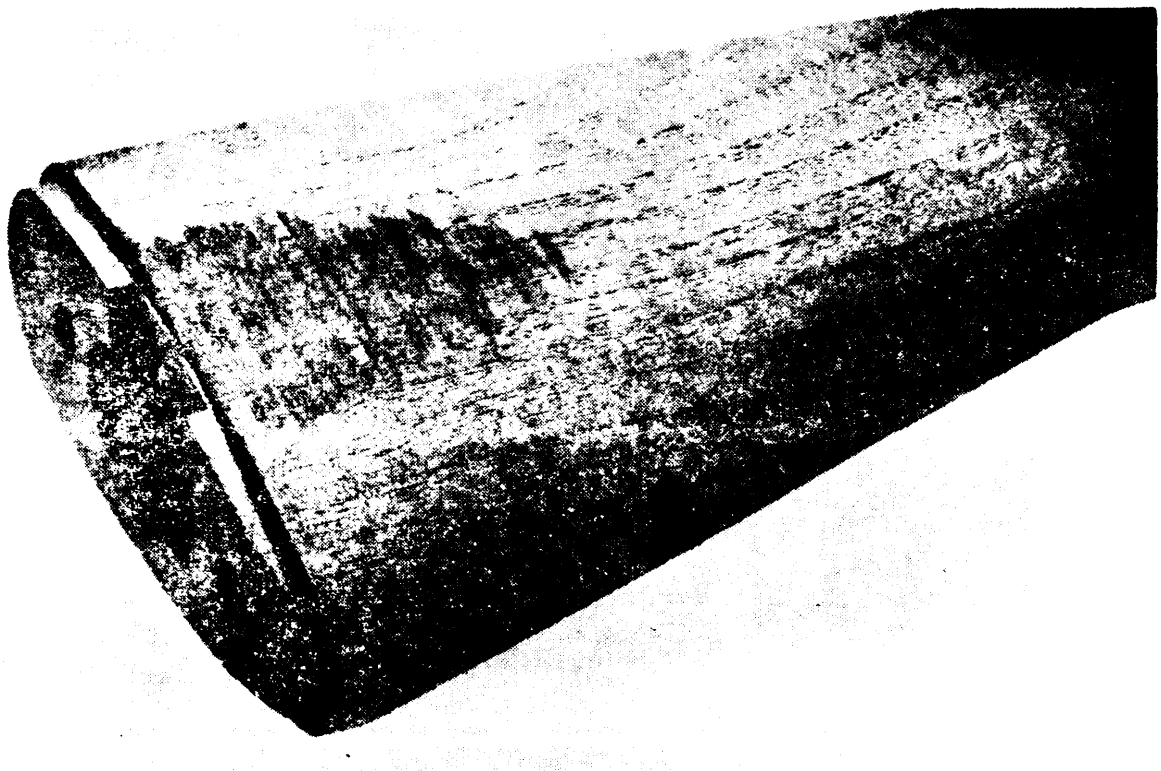


FIGURE 5. *Scant material defect.*

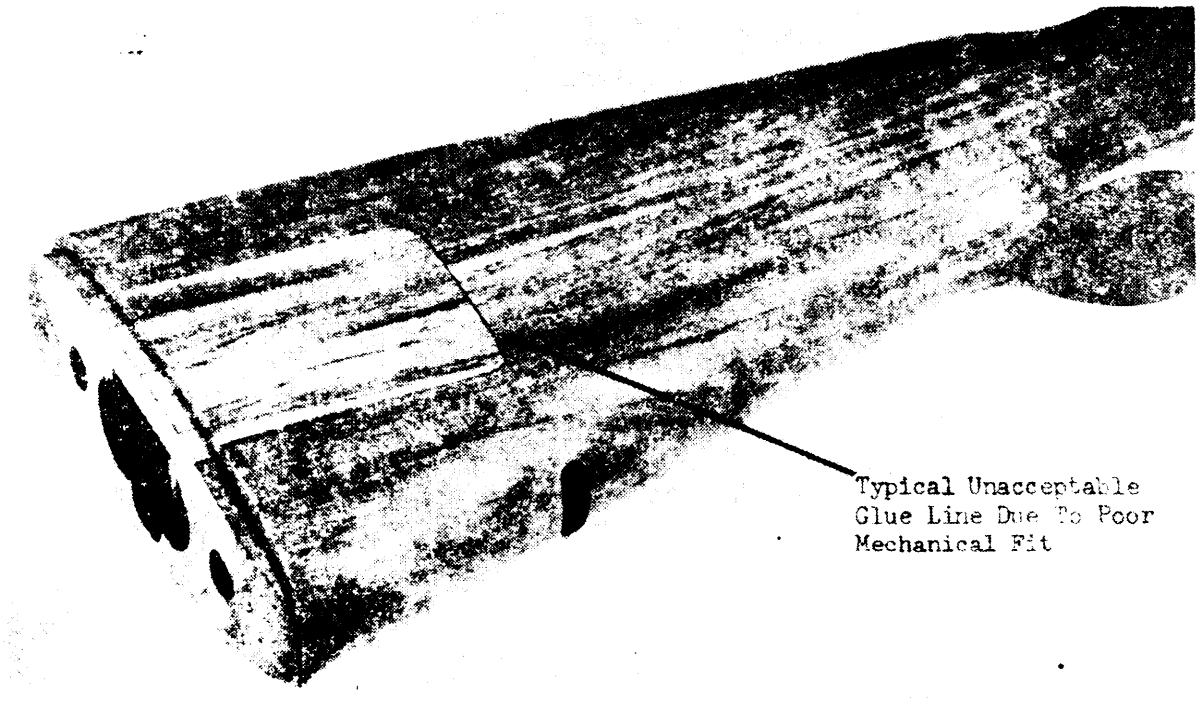


FIGURE 5A. *Scant material defect, patched.*

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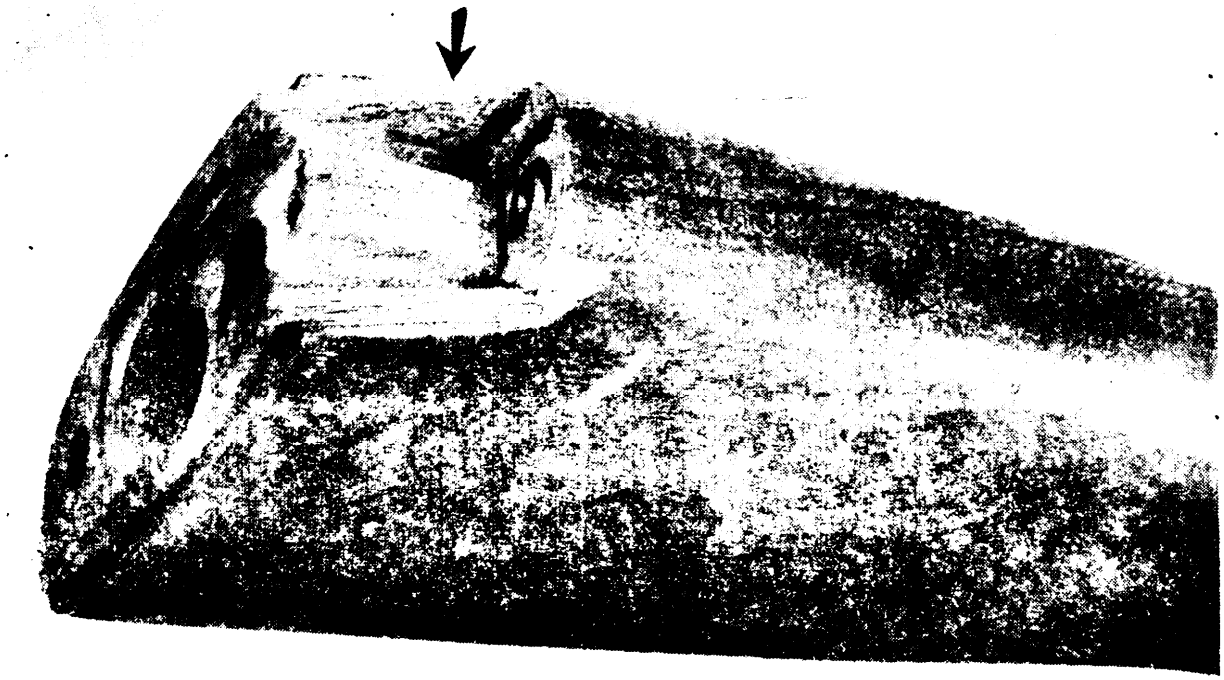


FIGURE 6. *Edge split defect.*

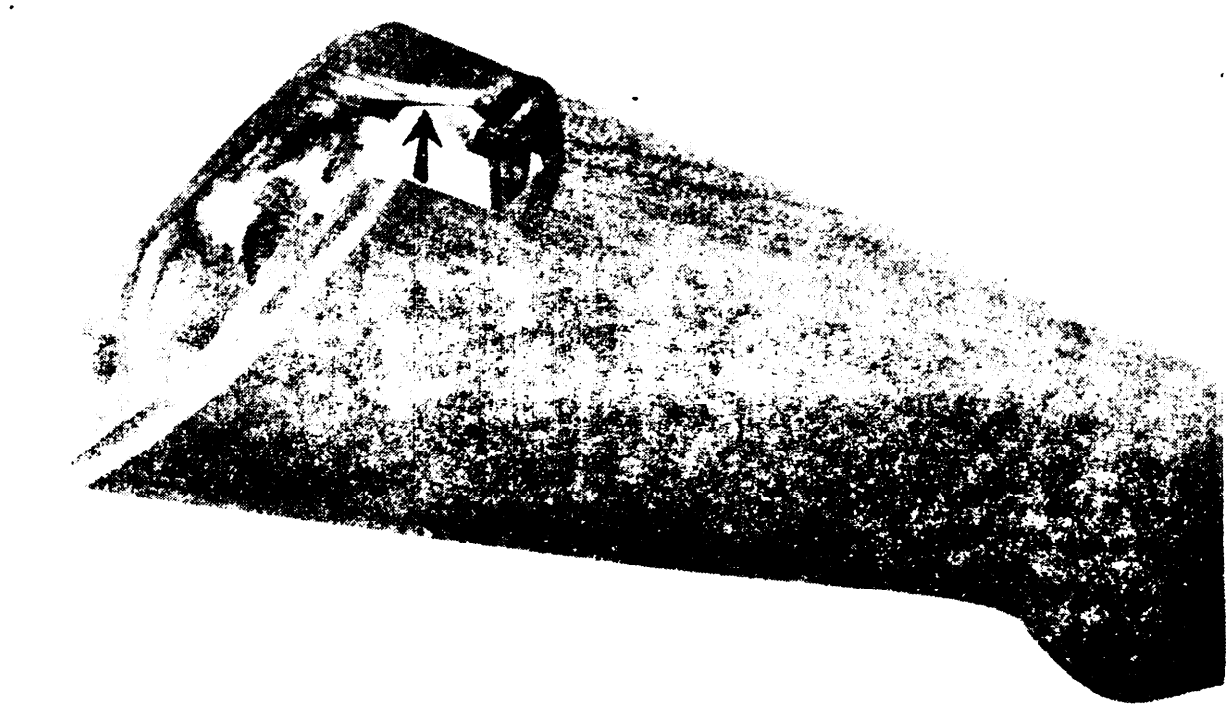


FIGURE 6A. *Edge split defect, patched.*

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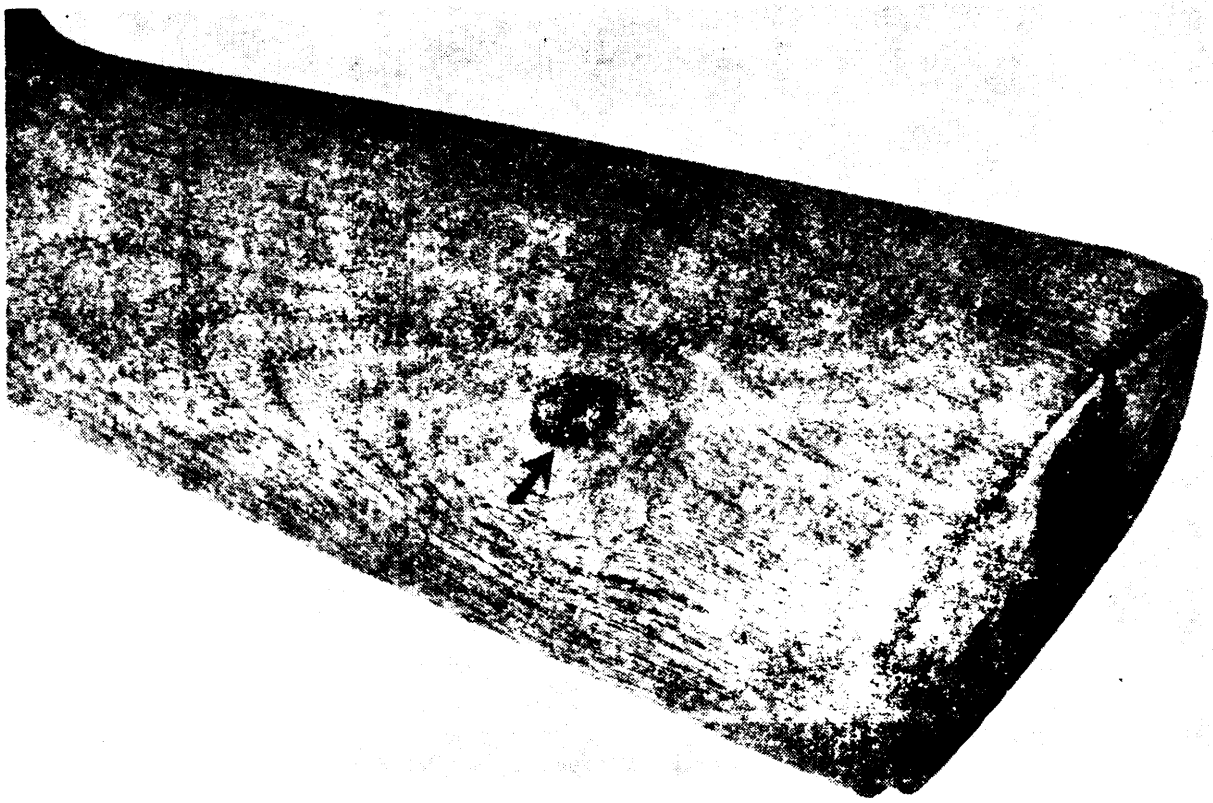


FIGURE 7. *Knothole defect.*



FIGURE 7A. *Knothole defect, patched.*

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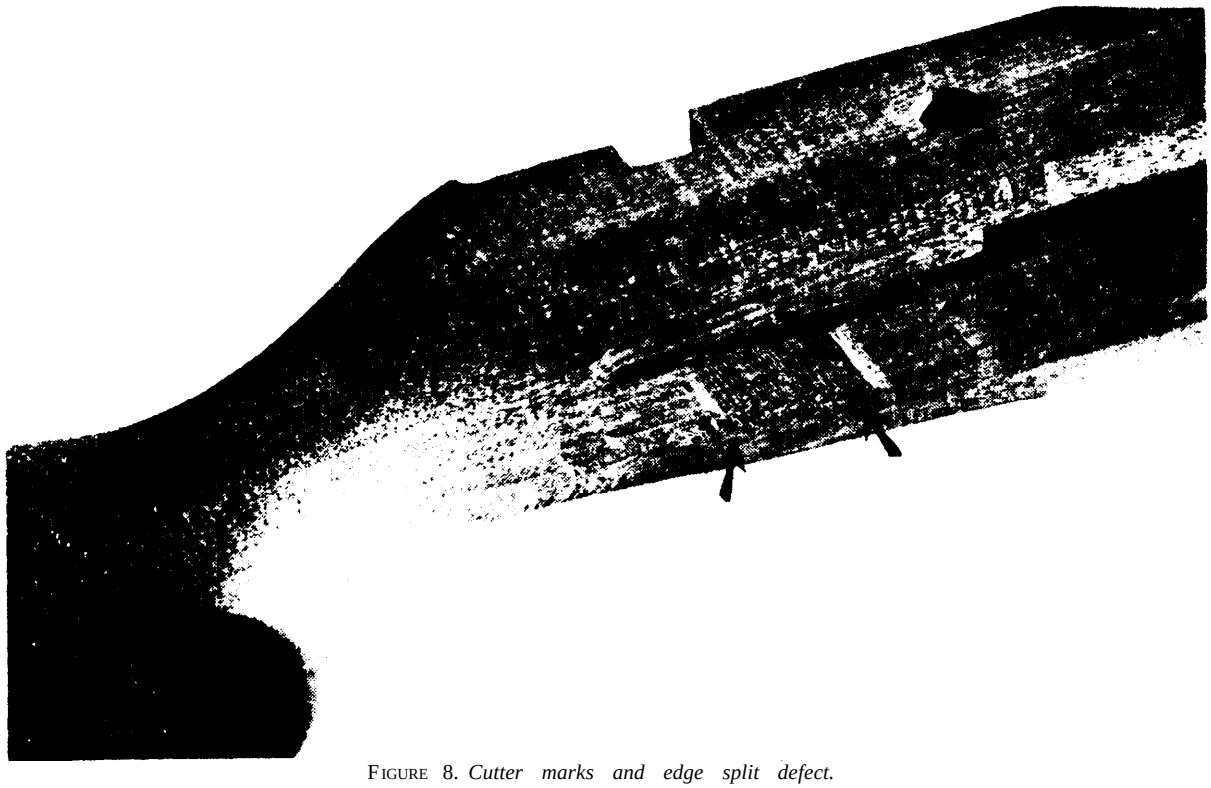


FIGURE 8. *Cutter marks and edge split defect.*



FIGURE 8A. *Cutter marks and edge split defect, patched.*

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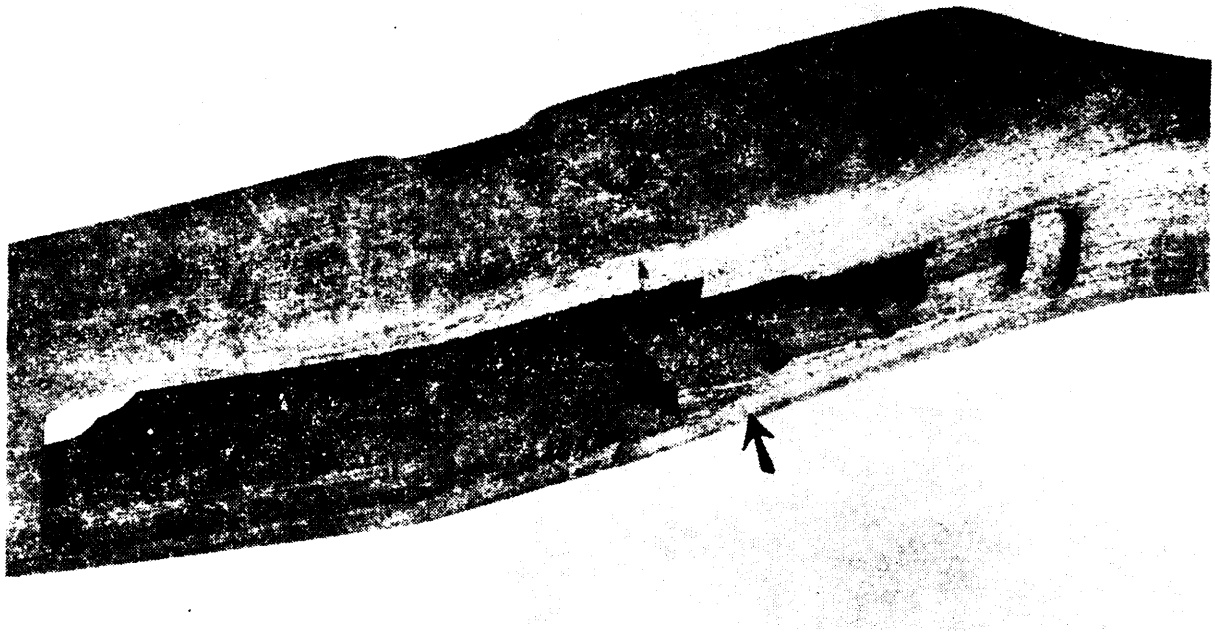


FIGURE 9. *Edge split defect.*

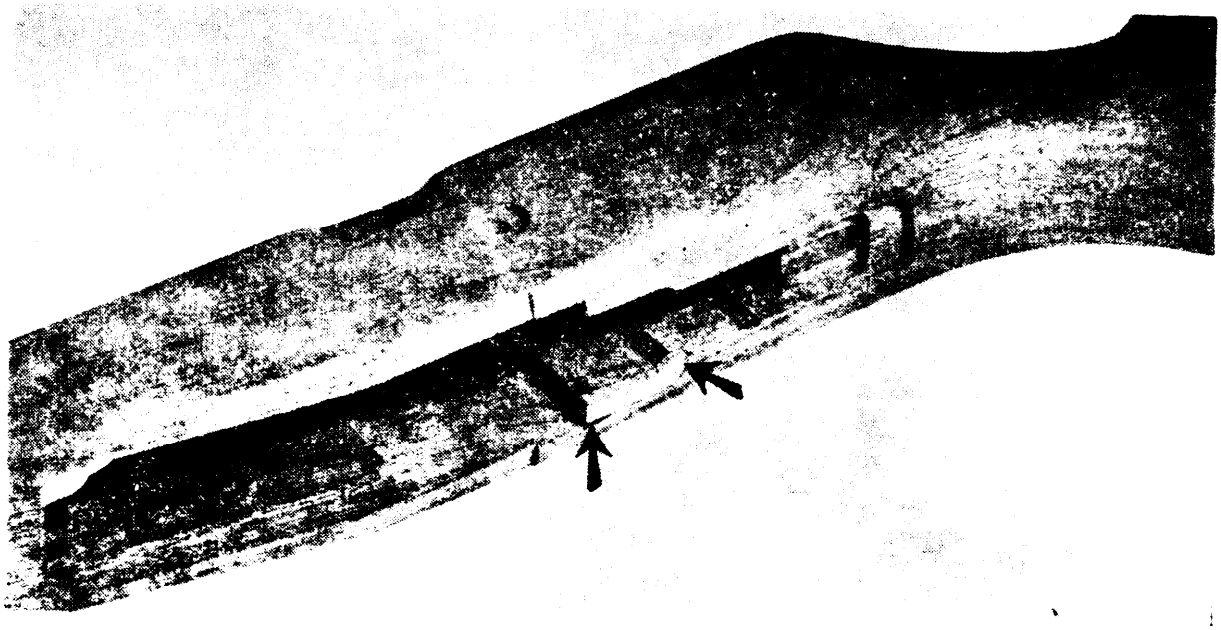


FIGURE 9A. *Edge split defect, patched.*

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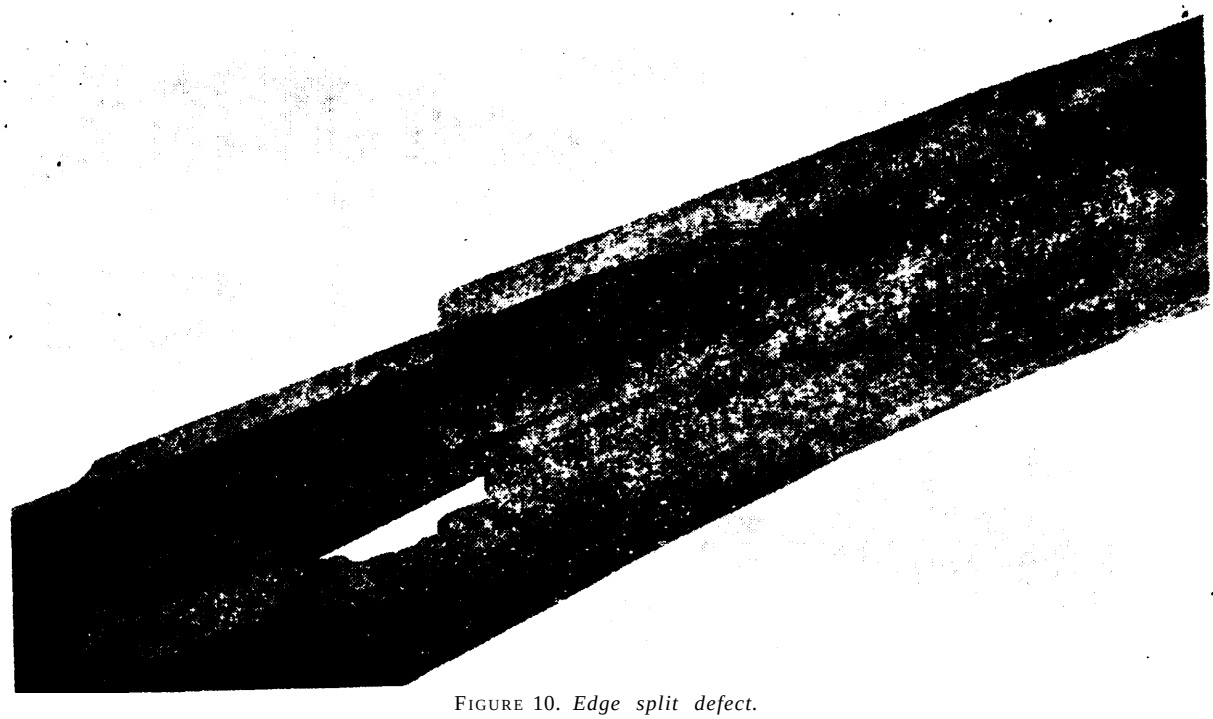


FIGURE 10. *Edge split defect.*

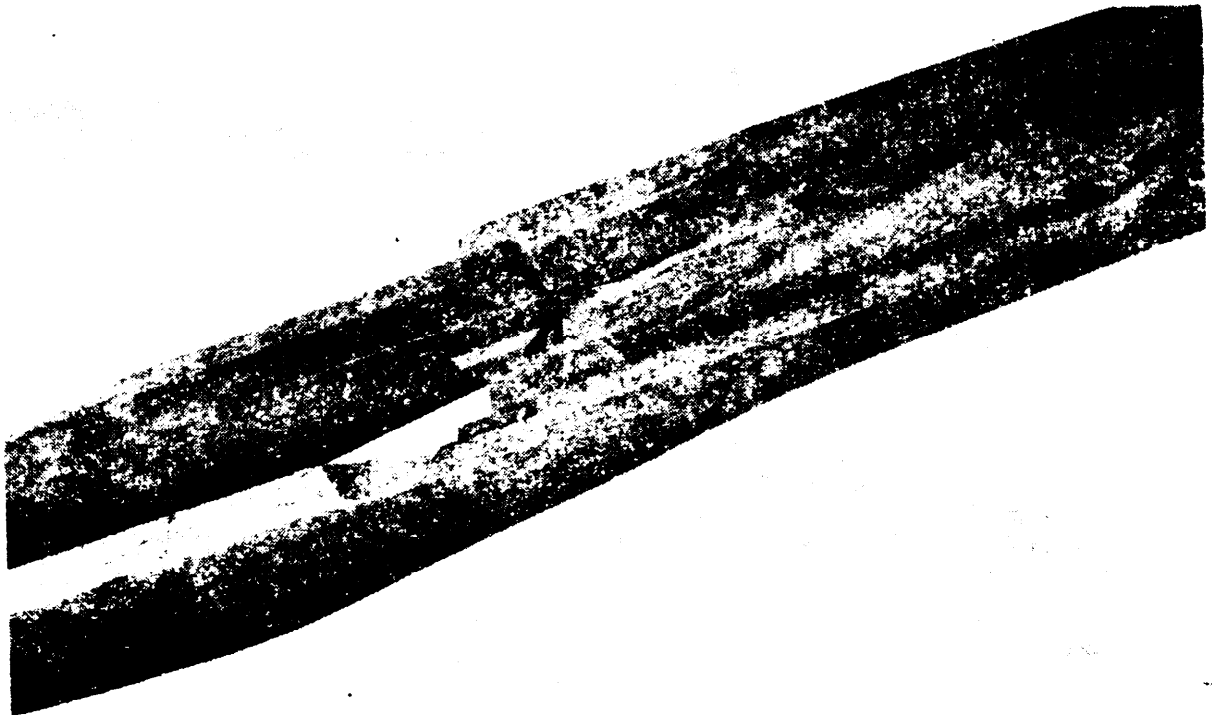


FIGURE 10A. *Edge split defect, patched.*

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FIGURE 11. *Edge split defect.*

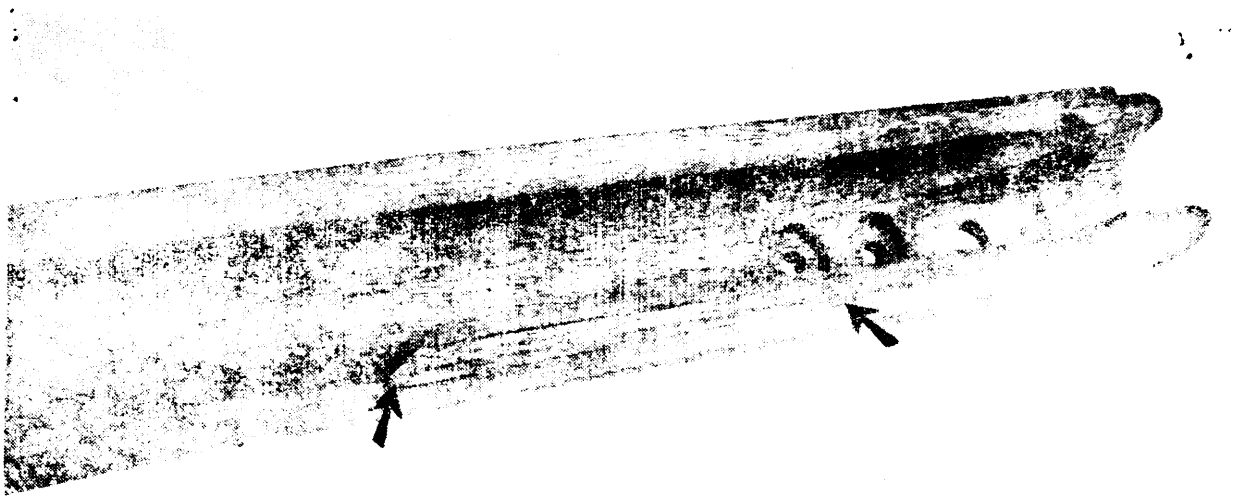


FIGURE 11A. *Edge split defect, patched.*

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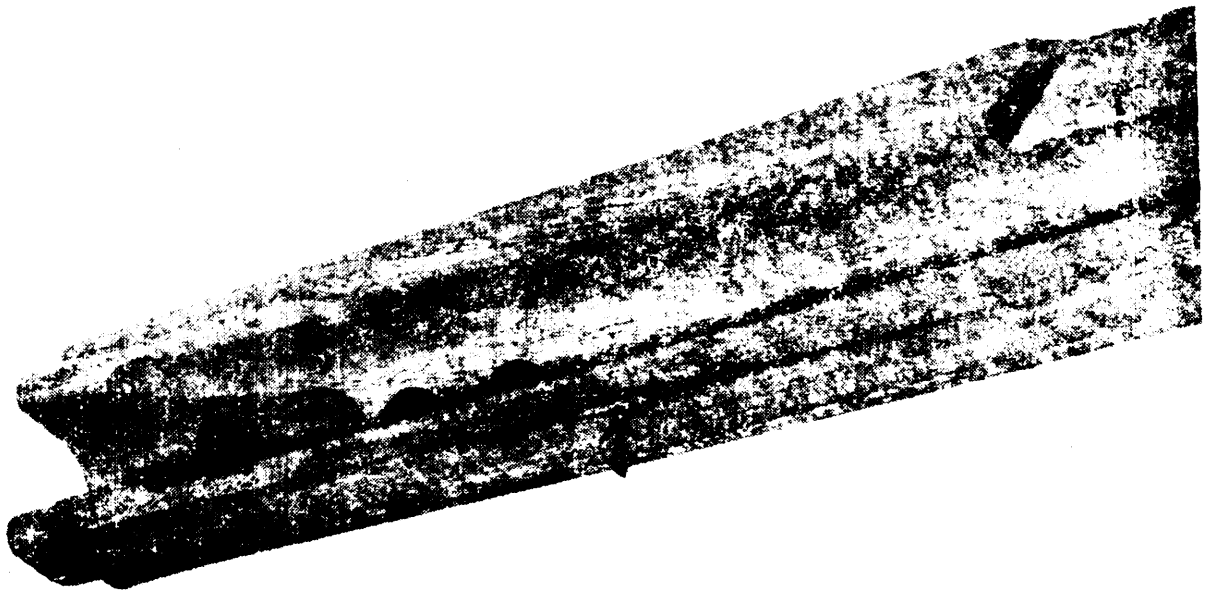


FIGURE 12. *Knot defect.*

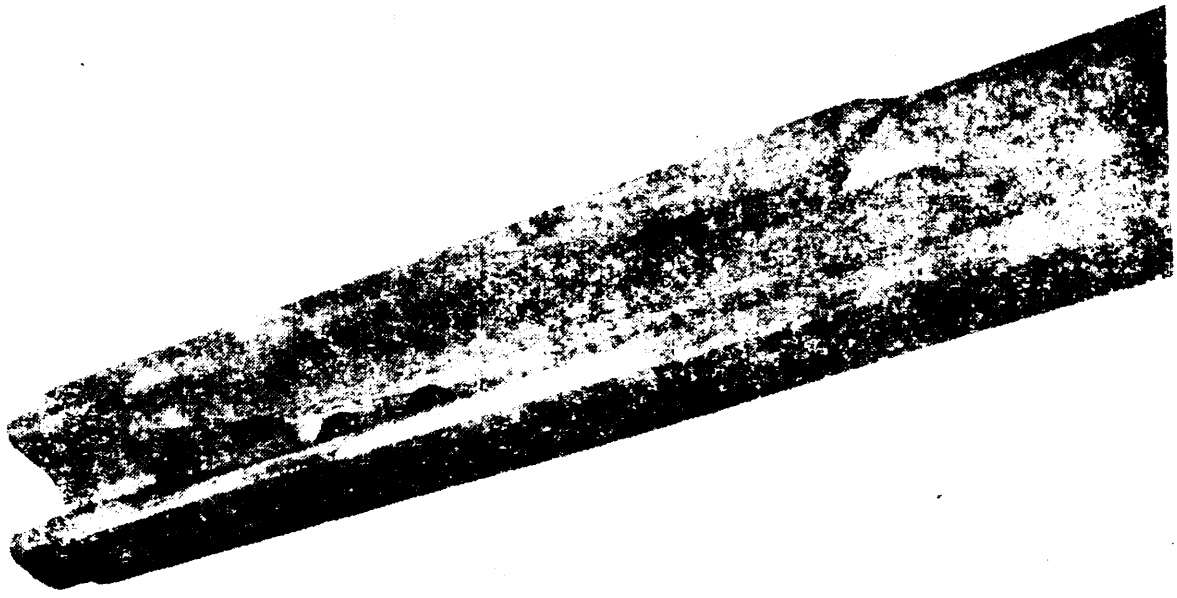


FIGURE 12A. *Knot defect, patched.*

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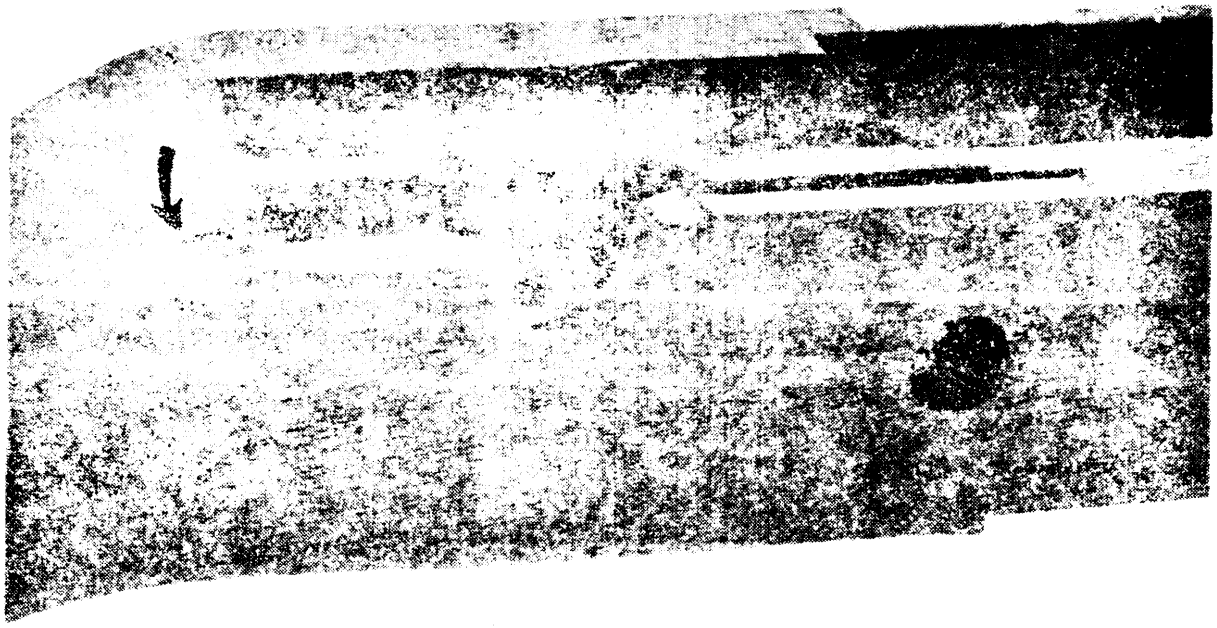


FIGURE 13. *Edge split defect.*

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FIGURE 13A. *Edge split defect patched.*

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FIGURE 14. *Corner split.*



FIGURE 14A. *Corner split, patched.*

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FIGURE 15. *Crack, front end.*

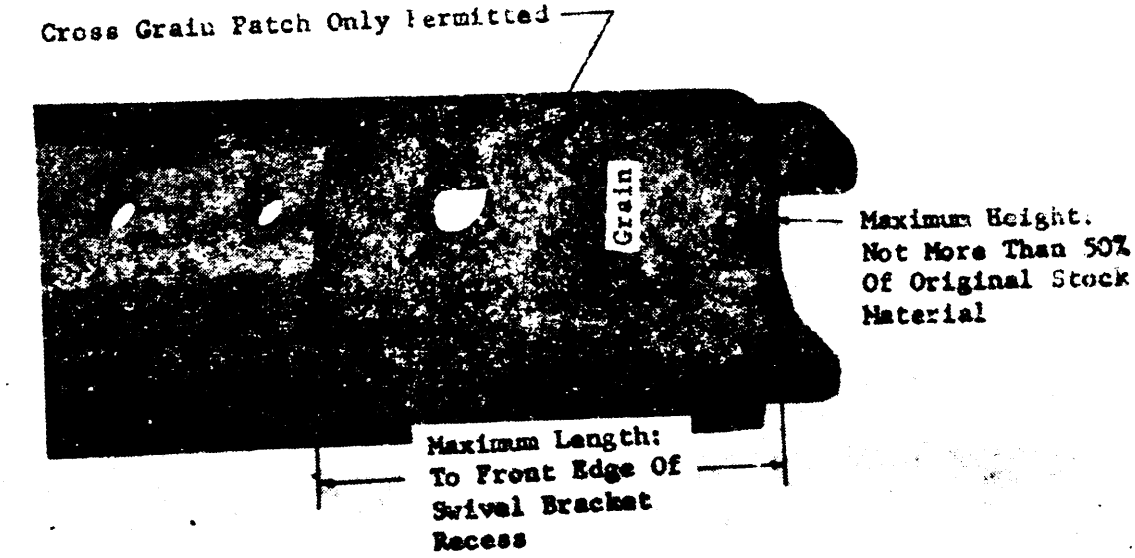


FIGURE 15A. *Crack front end patched.*

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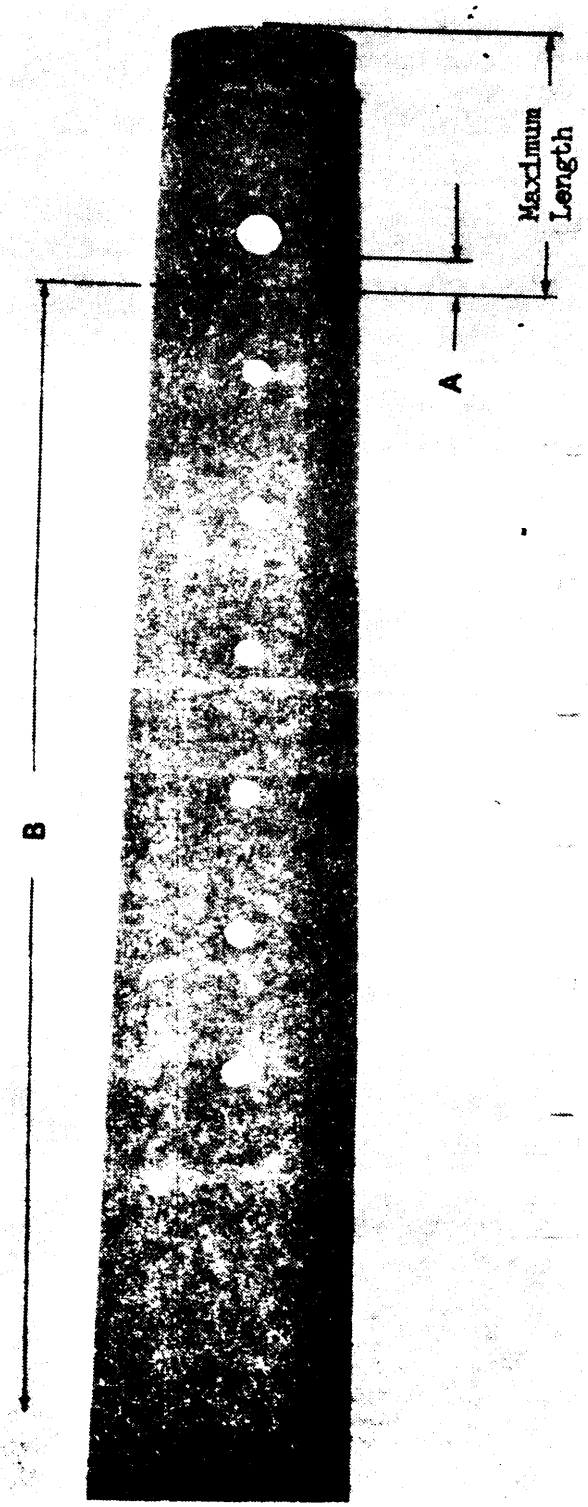


FIGURE 15B. *Front end and forearm area, M14B2.*

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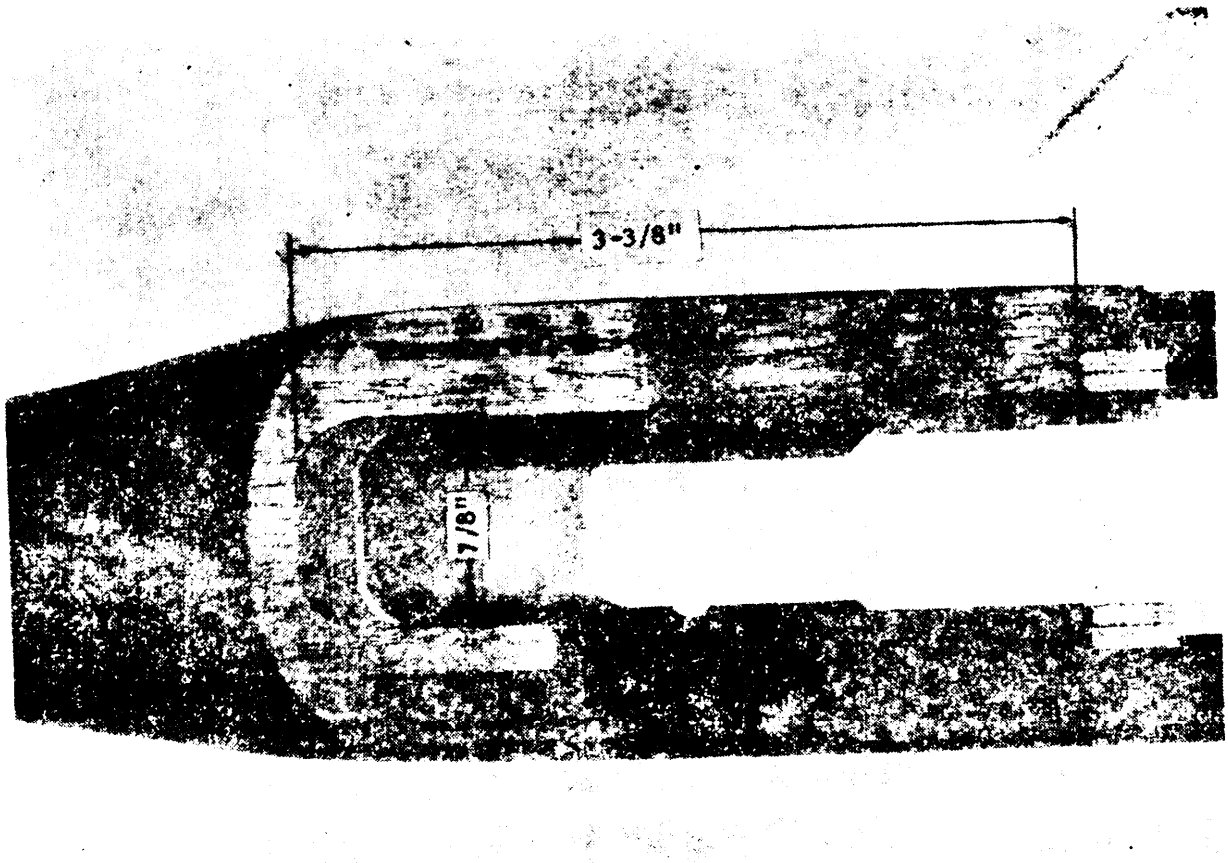


FIGURE 16. *Workmanship defect, patched.*



FIGURE 17. *Knothole defect.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:43Z
Check the source to verify that this is the current version before use.

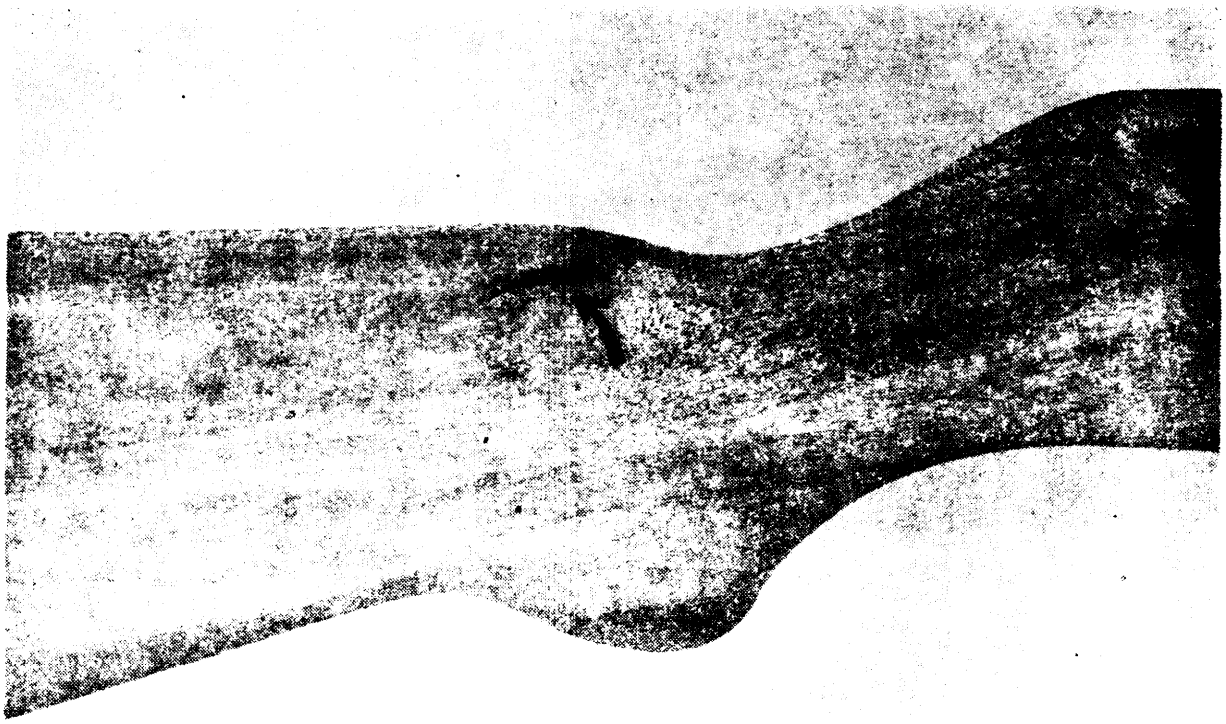


FIGURE 17A. *Knothole defect, filled.*

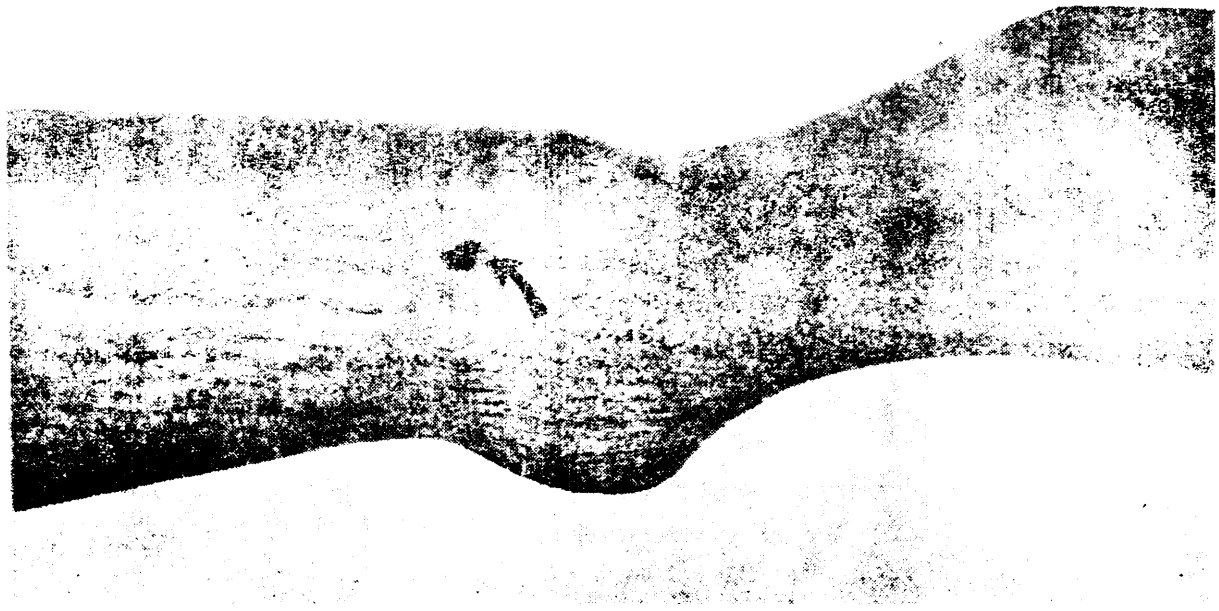


FIGURE 18. *Open defect.*

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FIGURE 18A. *Open defect, filled.*

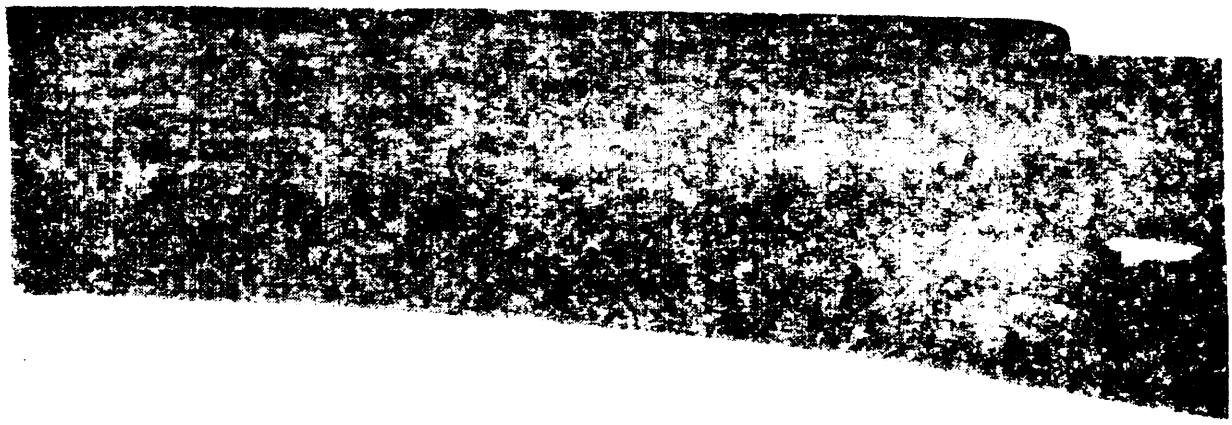


FIGURE 19. *Small knothole defect.*

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Check the source to verify that this is the current version before use.

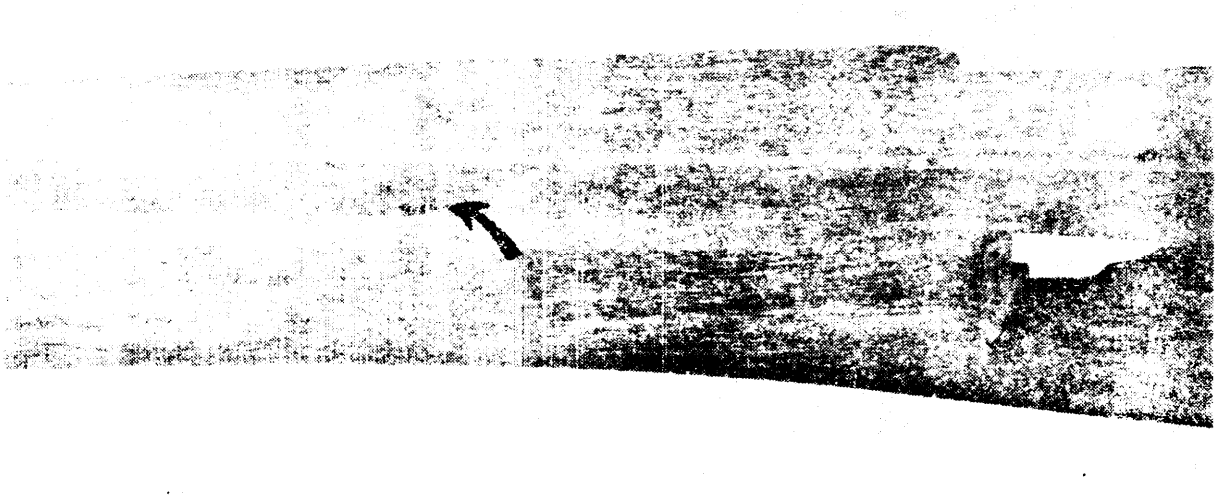


FIGURE 19A. *Small knothole defect, filled.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:43Z
Check the source to verify that this is the current version before use.

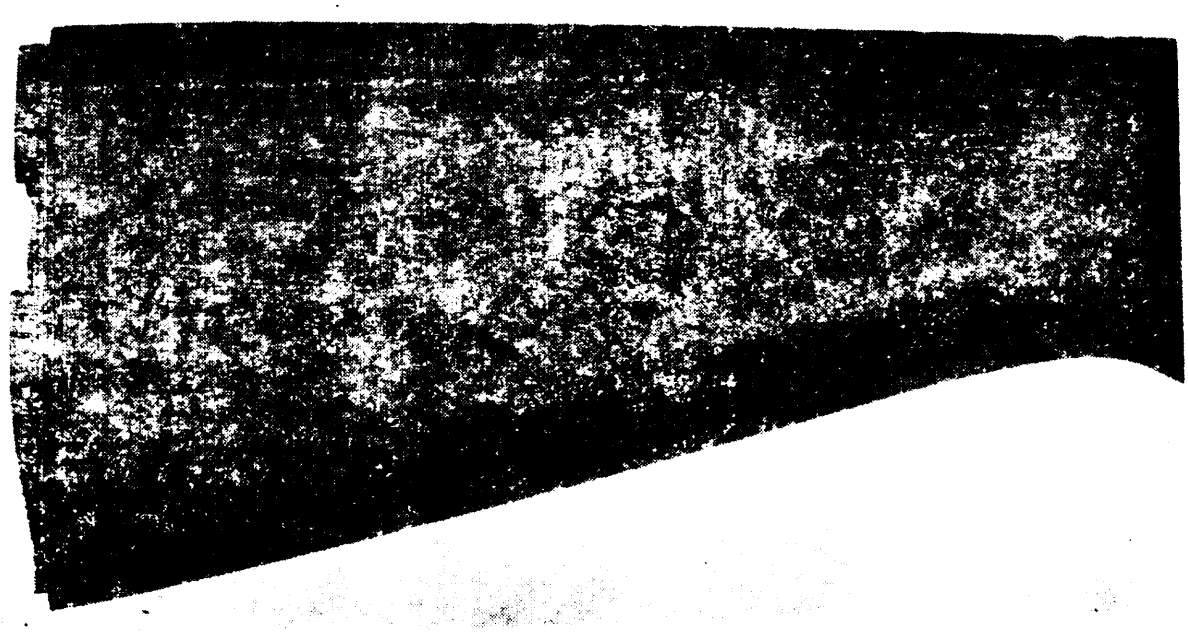


FIGURE 20. *Small knothole defect.*

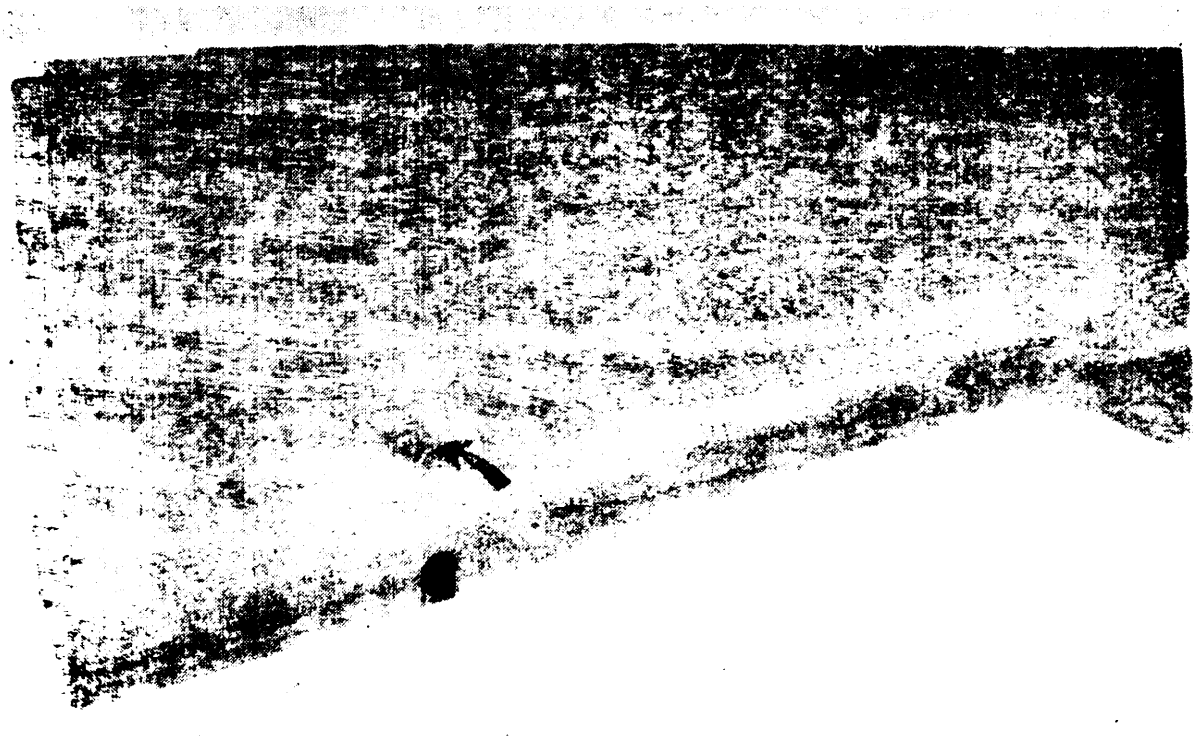


FIGURE 20A. *Small knothole defect, filled.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:43Z
Check the source to verify that this is the current version before use.

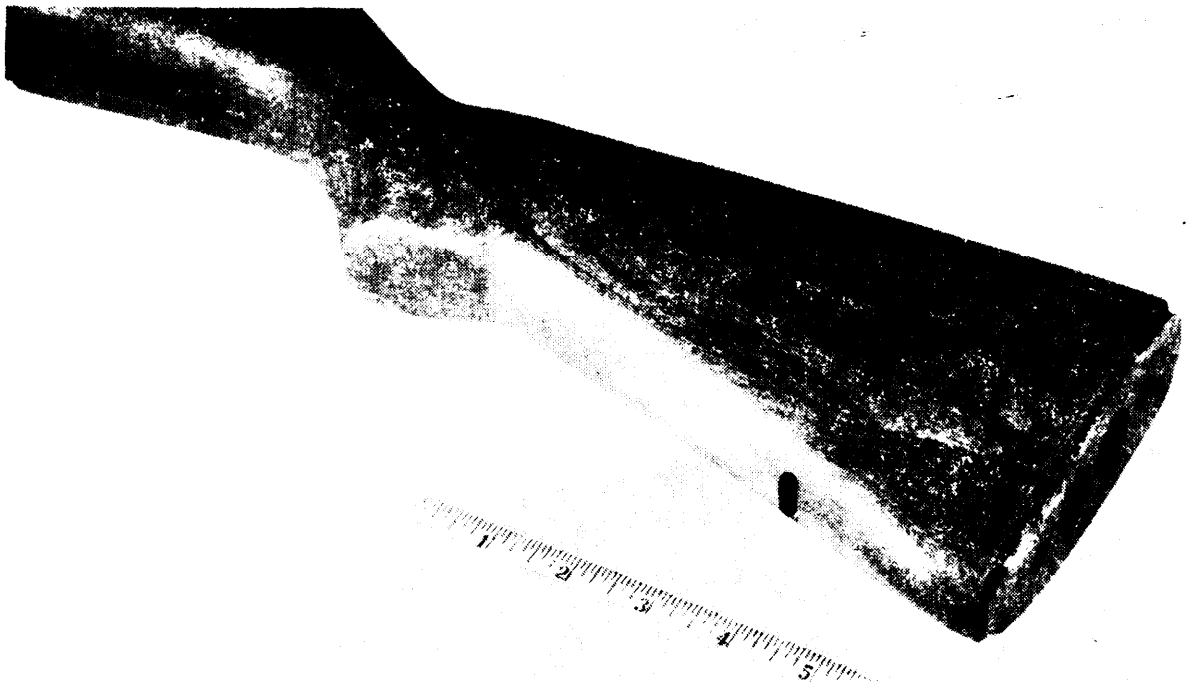


FIGURE 21. *Not acceptable patch.*

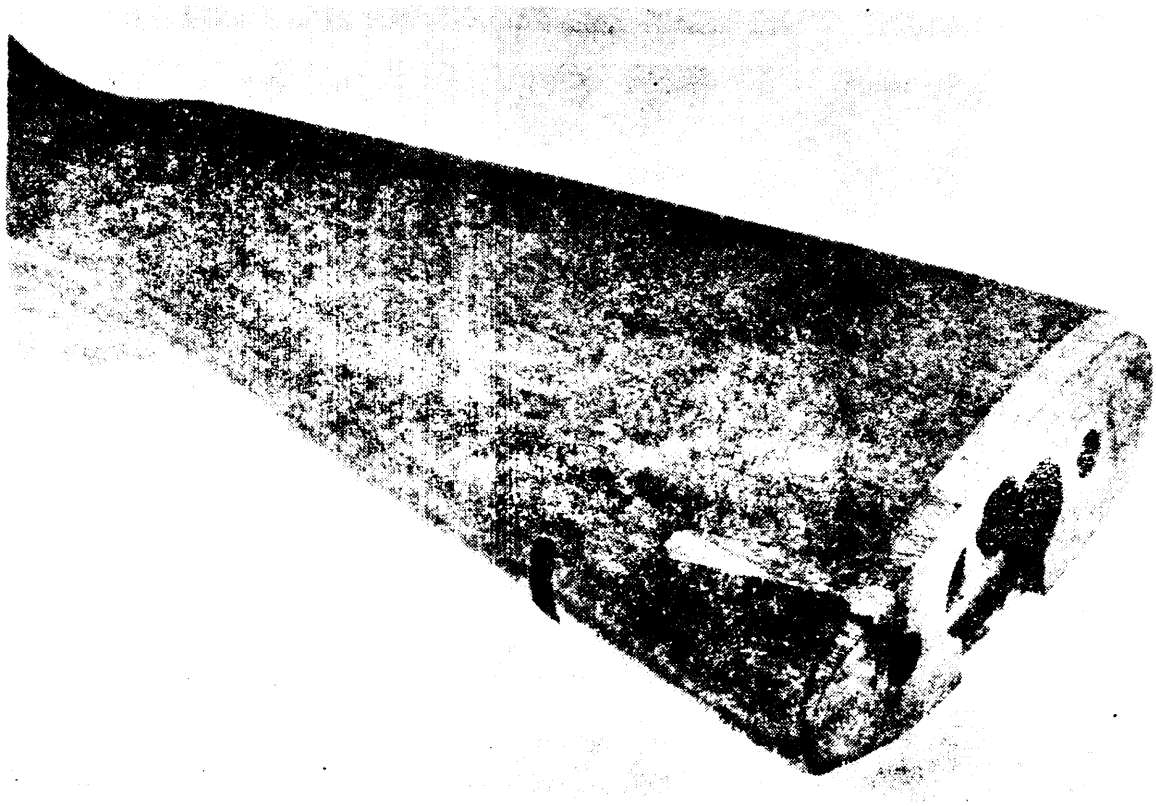


FIGURE *22. Not acceptable patch.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:43Z
Check the source to verify that this is the current version before use.

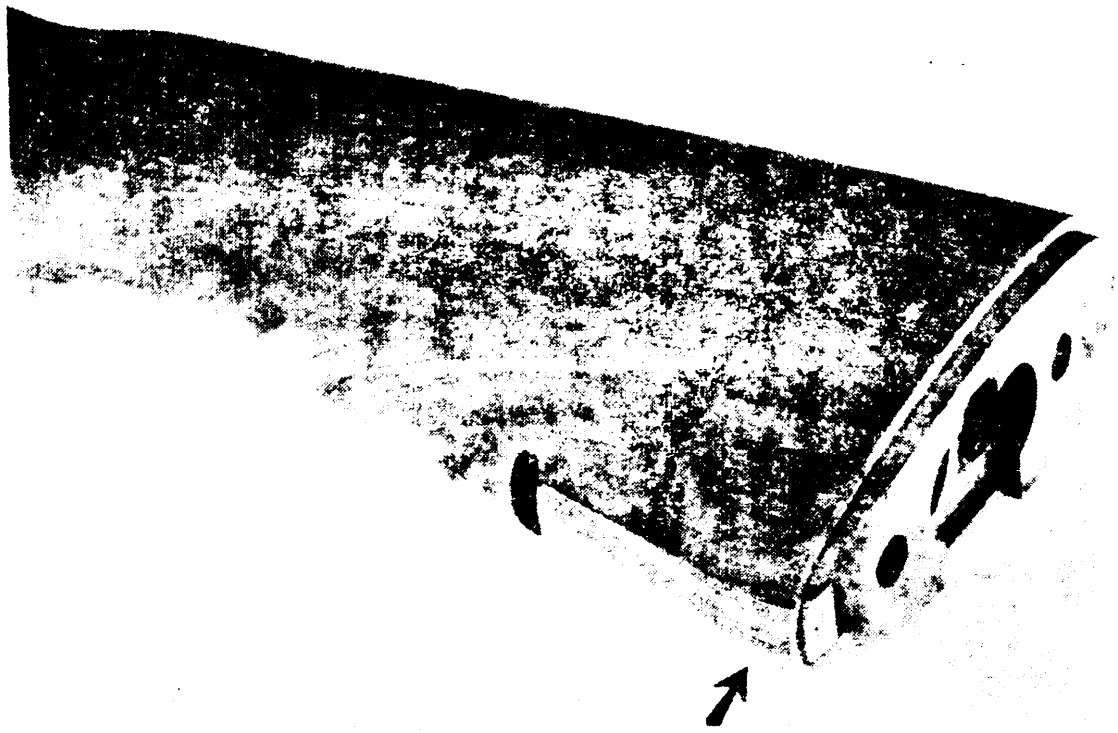


FIGURE 23. *Not acceptable patch.*

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Check the source to verify that this is the current version before use.

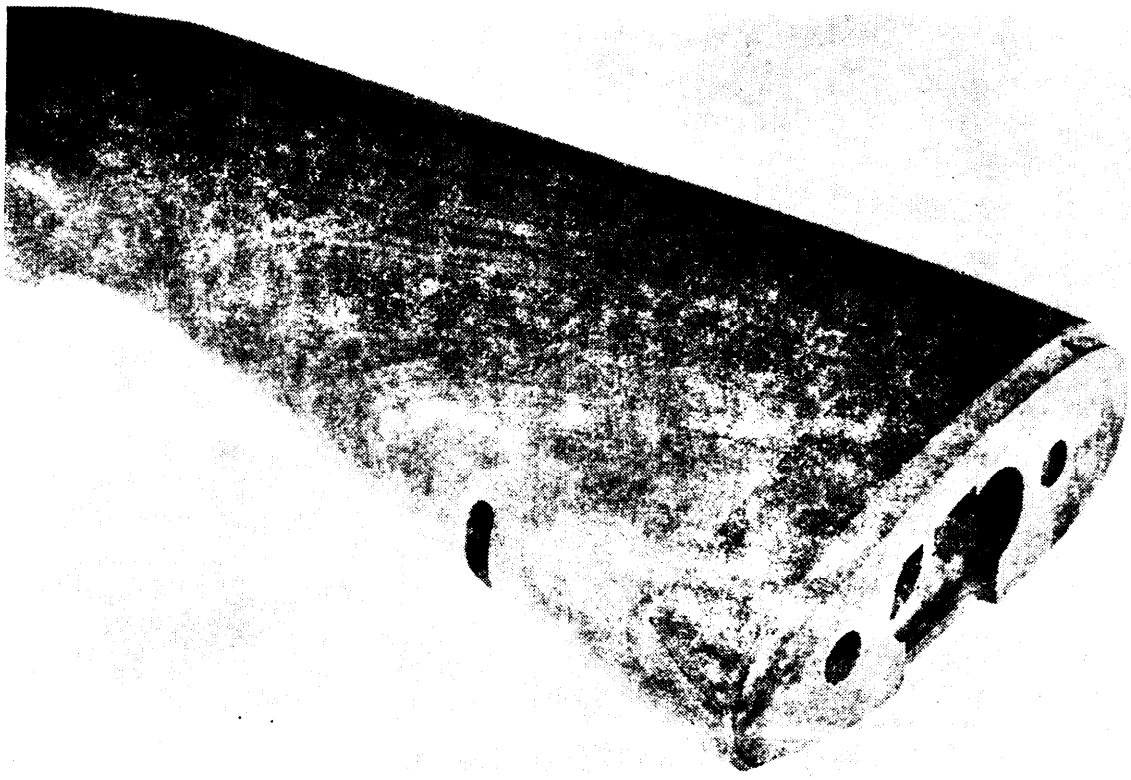


FIGURE 24. *Not acceptable patch.*

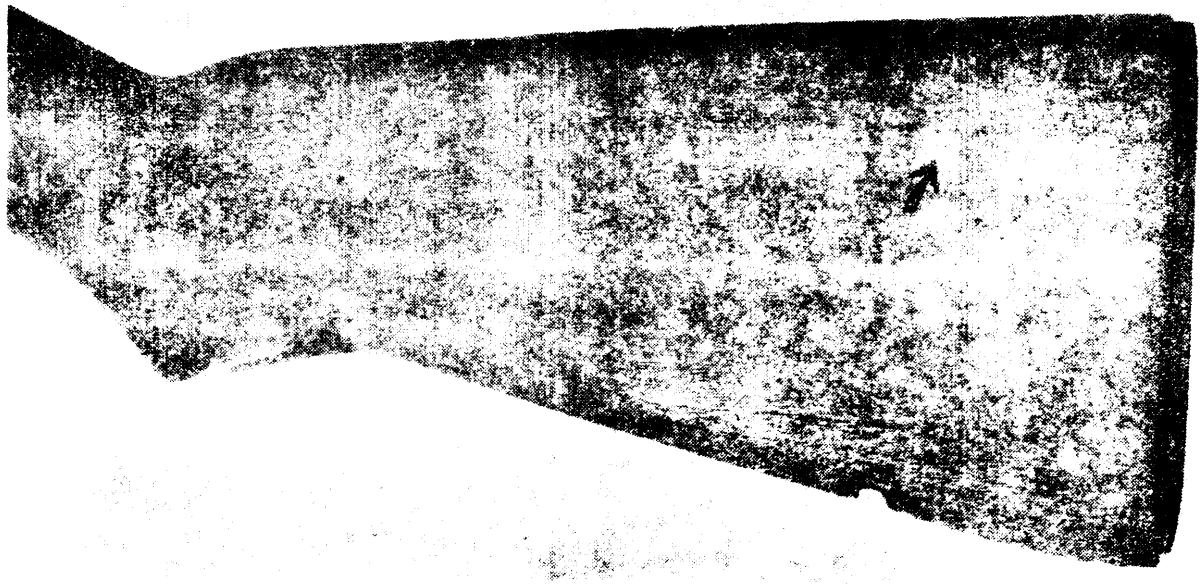


FIGURE 25. *Not acceptable patch.*

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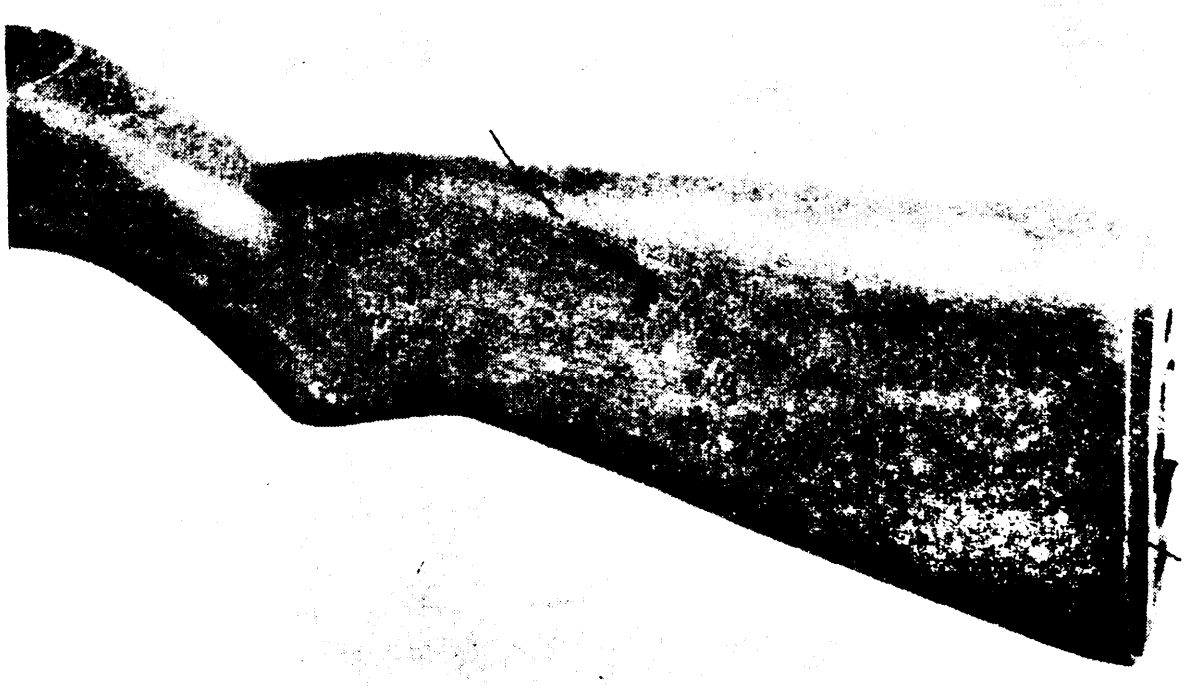


FIGURE 26. *Not acceptable patch.*

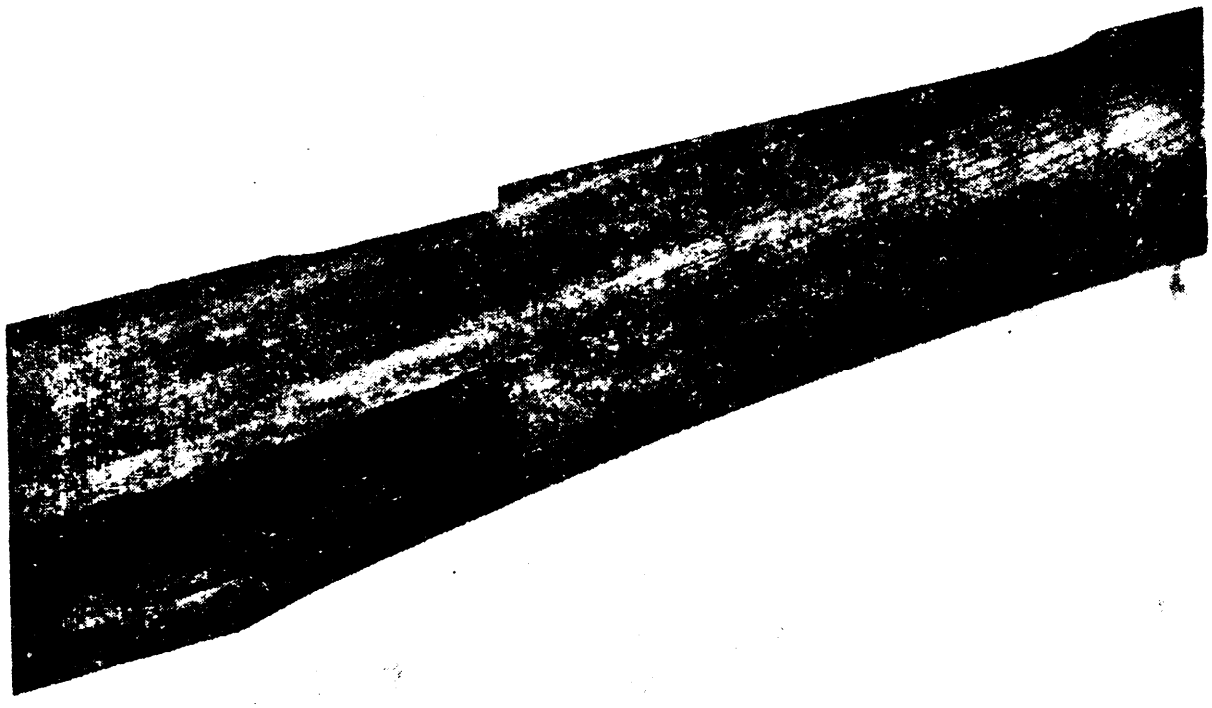


FIGURE 27. *Not acceptable patch.*

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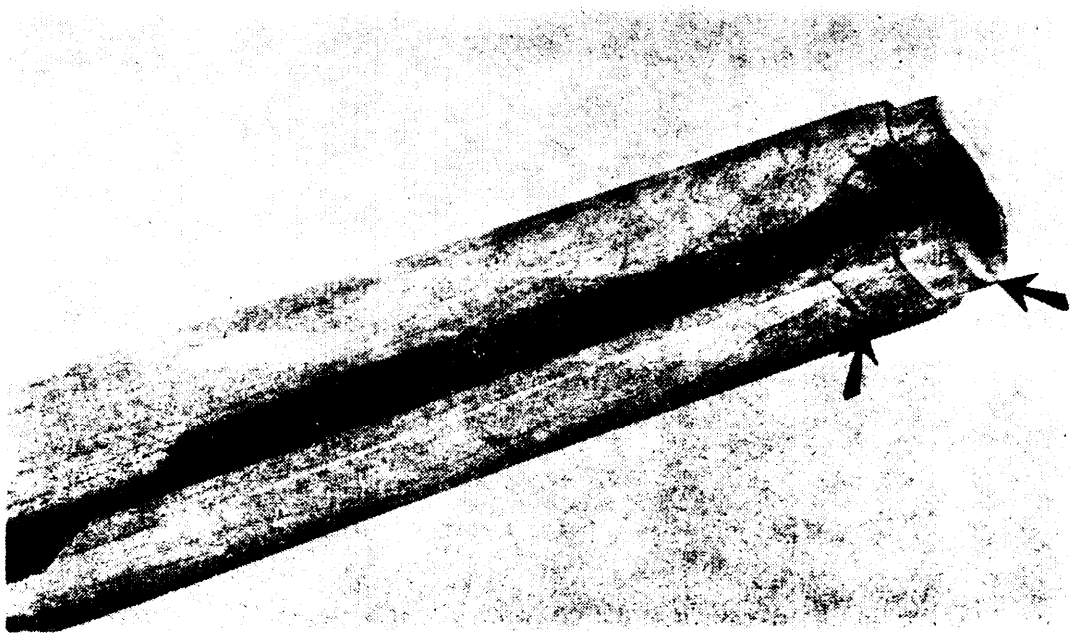


FIGURE 28. *Not acceptable patch.*

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